
APPLICATIONS OF STOCHASTIC CALCULUS FOR FINANCE

Lecture Notes

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STOCHASTIC CALCULUS



1 Lecture 1 - Stochastic Calculus I

$(\Omega, \mathcal{F}, (\mathcal{F})_{t \geq 0}, \mathbb{P})$ be a filtered probability space. A process M defined on it is a continuous martingale, if

- $t \mapsto M_t$ is continuous a.s
- $M_t \in \mathcal{L}'(\mathbf{P}) \quad \forall t$
- M is adapted to $(\mathcal{F})_t$
- $\mathbb{E}[M_t | \mathcal{F}_s] = M_s \quad \forall s < t$

M is a continuous local martingale if there exist stopping times $T_n, T_n \uparrow \infty$ a.s and $M^{T_n} 1_{(T_n > 0)}$ (where M^{T_n} is a stopped process) are bounded true martingales. ([1], page 121) Local Martingales are much more useful to work with as we will see later. The most important continuous martingales is **Brownian Motions**, which has independent, gaussian increments satisfying,

$$B_t - B_s \sim \mathcal{N}(0, t - s)$$

It can be described as a **centred Gaussian process** with $Cov(B_s, B_t) = s \quad \forall s \leq t$ (when $B_0 = 0$).

1.1 Stochastic Integrals

Suppose H, X processes, we want to define an integral which we write as,

$$(H \cdot X)_t = \int_0^t H_s dX_s = \lim_{\substack{|t_{i+1} - t_i| \rightarrow 0 \\ 0 < t_1 < \dots < t_n = t}} \sum_i \underbrace{H_{t_i}}_{\text{known at } t_i} \underbrace{(X_{t_{i+1}} - X_{t_i})}_{\text{future increment}}. \quad (1.1)$$

Condition on H : Predictable (or progressive) and X -integrable, which means being locally bounded is enough.

Condition on X : A continuous semi-martingale i.e. $X_t = X_0 + M_t + A_t$
 M_t is a Continuous Martingale, $M_0 = 0$ and A_t is a Continuous Finite Variation process with $A_0 = 0$. Such a decomposition is unique: There are no continuous martingales which are finite variation other than constants.

The most common semimartingales are Ito processes.

1.2 Ito's Process

$$X_t = X_0 + \int_0^t \sigma_s dB_s + \int_0^t \mu_s ds$$

With B_s is brownian motion, σ^2 is infinitesimal variance, μ is drift and both can depend on past satisfying,

$$\int_0^t (\sigma_s^2 + |\mu_s|) ds < \infty \quad \forall t$$

For "nice" σ and μ we expect:

$$\text{dist}(X_{t+h} - X_t | \mathcal{F}_t) \approx \mathcal{N}(\mu_t h, \sigma_t^2 h)$$

Important Notation: If $Y_t = \int_0^t H_s dX_s$ then $dY_t = H_t dX_t$

1.3 Some Properties of Stochastic Integrals

We often write

$$dY_t = H_t dX_t.$$

The stochastic integral is linear in both the integrand and the integrator. In particular,

$$(aH_t + bK_t) dX_t = aH_t dX_t + bK_t dX_t,$$

and

$$H_t d(aX_t + bY_t) = aH_t dX_t + bH_t dY_t.$$

Moreover, the stochastic integral satisfies an associativity property. If

$$dY_t = K_t dX_t \quad \text{and} \quad dZ_t = H_t dY_t,$$

then

$$dZ_t = (H_t K_t) dX_t.$$

1.4 Preservation of the Local Martingale Property

Most importantly, stochastic integrals preserve the local martingale property. If M is a local martingale and

$$dN_t = H_t dM_t,$$

then N is also a local martingale.

This can be seen by viewing the stochastic integral as the limit of Riemann sums. For a partition $0 = t_0 < t_1 < \dots < t_n = t$, we consider

$$\sum_{i=1}^n H_{t_i} (X_{t_{i+1}} - X_{t_i}).$$

Since M is a martingale, we have

$$\mathbb{E}[H_{t_i} (X_{t_{i+1}} - X_{t_i}) | \mathcal{F}_{t_i}] = 0.$$

Let $s = t_j$. Then

$$\mathbb{E} \left[\sum_{i=1}^n H_{t_i} (X_{t_{i+1}} - X_{t_i}) \middle| \mathcal{F}_s \right] = \sum_{i=1}^j H_{t_i} (X_{t_{i+1}} - X_{t_i}).$$

2 Lecture 2 - Stochastic Calculus II: quadratic variation and Ito's formula

2.1 Finite Variation

A path $X : [0, T] \rightarrow \mathbb{R}$ is said to have *finite (or bounded) variation* if

$$\sup_{0=t_0 \leq t_1 \leq \dots \leq t_n=T} \sum_{i=0}^{n-1} |X(t_{i+1}) - X(t_i)| < \infty.$$

This quantity represents the *total amount of up-and-down movement* of the path.

Any differentiable path has finite variation.

Moreover, any finite variation path can be written as the difference of two increasing functions.

Consequently, the Riemann–Stieltjes integral is well-defined and

$$\int_0^T H(s) dX(s) = \int_0^T H(s) \dot{X}(s) ds,$$

if

$$dX_t = \dot{X}_t dt,$$

that is, if X is differentiable. **Brownian paths are not of finite variation.**

In fact, if one zooms in by scaling time and space equally, the oscillations become larger.

However, Brownian motion has finite quadratic variation. In particular,

$$\lim_{|t_{i+1}-t_i| \rightarrow 0} \sum_{i=0}^{n-1} (B(t_{i+1}) - B(t_i))^2 = T, \quad 0 = t_0 < \dots < t_n = T \in (0, \infty),$$

almost surely. i.e. the quadratic variation of Brownian motion on $[0, T]$ is equal to $T \in (0, \infty)$

almost surely. This is easy to check by considering the mean and variance of the sum

This has profound consequences for calculus. If X is differentiable then:

$$\begin{aligned} X(T)^2 - X(0)^2 &= \int_0^T \frac{d}{dt}(X(t)^2) dt = 2 \int_0^T X(t) \dot{X}(t) dt \\ &= 2 \int_0^T X(t) dX(t). \end{aligned}$$

But for a Brownian path:

$$\begin{aligned} B(T)^2 - B(0)^2 &= \sum_i (B(t_{i+1})^2 - B(t_i)^2) \\ &= \sum_i (B(t_{i+1}) + B(t_i)) (B(t_{i+1}) - B(t_i)) \\ &= 2 \sum_i B(t_i) (B(t_{i+1}) - B(t_i)) + \sum_i (B(t_{i+1}) - B(t_i))^2 \\ &= 2 \int_0^T B(t) dB(t) + \underbrace{T}_{\text{Itô correction}}. \end{aligned}$$

For semimartingales X, Y we can define quadratic variation / covariation:

$$\langle X \rangle(t) = \lim \sum_i (X(t_{i+1}) - X(t_i))^2$$

$$\langle X, Y \rangle(t) = \lim \sum_i (X(t_{i+1}) - X(t_i))(Y(t_{i+1}) - Y(t_i))$$

These are always finite variation processes themselves. For Itô processes

$$dX_t = \sigma_t dB_t + \mu_t dt, \quad dY_t = \sigma'_t dB'_t + \mu'_t dt$$

$$\langle X \rangle(t) = \int_0^t \sigma_s^2 ds \quad \langle X, Y \rangle(t) = \int_0^t \sigma_s \sigma'_s \rho_s ds$$

where ρ is the correlation between the BMs B and B' which may depend on time. We often write

$$d\langle X, Y \rangle = dX dY$$

with the rules:

$$(dt)^2 = 0, \quad dt dB_t = 0,$$

$$(dB_t)^2 = dt, \quad dB_t dB'_t = \begin{cases} \rho dt, & \text{if } \text{Corr}(B_t, B'_t) = \rho, \\ 0, & \text{if } B \text{ and } B' \text{ are independent.} \end{cases}$$

Itô's formula generalizes the calculation we did for the square of BM: if $f \in C^2(\mathbb{R})$ then

$$f(X_t) - f(X_0) = \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d\langle X \rangle_s$$

extends to multidimensional case (Theorem 3.3, Chapter IV of [1])

$$X = (X^1, \dots, X^d) \quad \text{and} \quad F \in C^2(\mathbb{R}^d, \mathbb{R})$$

Then

$$F(X_t) - F(X_0) = \sum_i \int_0^t \frac{\partial F}{\partial x_i}(X_s) dX_s^i + \frac{1}{2} \sum_{i,j} \int_0^t \frac{\partial^2 F}{\partial x_i \partial x_j}(X_s) d\langle X^i, X^j \rangle_s$$

Special cases:

- $f(X_t, t)$ (function of time + space)
- $X_t Y_t$ (product of two semimartingales)

Finally we mention Lévy's characterization of BM: If X is a continuous local martingale, then the following are equivalent

$$\langle X \rangle(t) = t \quad \Longleftrightarrow \quad X \text{ is a BM.}$$

(Theorem 3.6, Chapter IV of [1])

3 Lecture 3 - Stochastic Calculus III: SDEs and diffusions

3.1 Markov Process

A process X has the **(strong) Markov property** (w.r.t. the filtration $(\mathcal{F}_t)_{t \geq 0}$) if for each (stopping) time T the conditional distribution of $(X_{T+h}; h \geq 0)$ given $\mathcal{F}_T$ depends only on X_T . So if F is an integrable function of paths then

$$\mathbb{E}[F(X_{T+h}; h \geq 0) \mid \mathcal{F}_T] = G(X_T),$$

for some G (depending on F and T in general).

A Markov process is **time-homogeneous** if the conditional distribution of $(X_{T+h}, h \geq 0)$ given $X_T = x$ is that of the process started from x at time zero, so then

$$G(x) = \mathbb{E}_x[F(X_s, s \geq 0)]$$

↙ denotes X starts from x

A **diffusion** is an $\mathbb{R}^d$ -valued continuous process with the **(strong) Markov property**.

3.2 Kolmogorov's Backward Equation

Kolmogorov (1931) specified a diffusion by its infinitesimal mean (drift) and variance

$$\mathbb{E}[X_{t+h} - X_t \mid X_t = x] = b(t, x) h + o(h)$$

$$\mathbb{E}[(X_{t+h} - X_t)^2 \mid X_t = x] = a(t, x) h + o(h)$$

When $d = 1$. If $d > 1$ then b becomes a vector and a becomes a symmetric matrix. The transition densities

$$\mathbb{E}[X_t \in dy \mid X_s = x] = p_{s,t}(x, y) dy$$

satisfy a PDE with coefficients given by a and b .

Kolmogorov's approach was to solve the PDE and construct X (actually $\mathbb{P}$) from the transition densities.

Itô (1942): X can be constructed from a Brownian motion by solving an SDE

$$X_t = x + \int_0^t \sigma(s, X_s) dB_s + \int_0^t b(s, X_s) ds$$

where

$$(\sigma(s, x))^2 = a(s, x).$$

- Notice how we may interpret the drift and infinitesimal variance as describing “the way in which X behaves depends just on where it currently is” (in a time-homogeneous case where a, b don't depend on time).
- There is a large theory on existence and uniqueness for SDEs (Chapter IV of [1]). Solutions are rarely given by explicit formulae. We will just assume these properties for the SDEs we meet.
- By applying Itô's formula we can connect the SDE to Kolmogorov's PDE - **key technique!**

Consider

$$\mathbb{E}[h(X_T) \mid \mathcal{F}_t] = u(t, X_t) \quad (\text{by Markov property}).$$

Apply Itô's formula assuming u is smooth enough.

(4)

$$\begin{aligned} u(t, X_t) &= u(0, X_0) + \int_0^t \frac{\partial u}{\partial t}(s, X_s) ds + \int_0^t \frac{\partial u}{\partial x}(s, X_s) dX_s + \frac{1}{2} \int_0^t \frac{\partial^2 u}{\partial x^2}(s, X_s) d\langle X \rangle_s \\ &= \text{local martingale} + \int_0^t \left\{ \frac{\partial u}{\partial s}(s, X_s) + b(s, X_s) \frac{\partial u}{\partial x}(s, X_s) + \frac{1}{2} a(s, X_s) \frac{\partial^2 u}{\partial x^2}(s, X_s) \right\} ds \end{aligned}$$

Now by the tower property $u(t, X_t)$ is a martingale so **the drift term must be zero**. So u solves the p.d.e.

$$\underbrace{\frac{\partial u}{\partial t} + b(t, x) \frac{\partial u}{\partial x} + \frac{1}{2} a(t, x) \frac{\partial^2 u}{\partial x^2}}_{\mathcal{L} \text{ "the generator"}} = 0$$

By taking h to be a “delta function” we find the transition density itself solves the p.d.e. For fixed T, y

$$\frac{\partial}{\partial t} p_{t,T}(x, y) + \mathcal{L} p_{t,T}(x, y) = 0.$$

This is **Kolmogorov's backward equation**.

4 Lecture 4 - Stochastic Calculus IV: Martingales

4.1 Coin-Tossing Model

Let $(\varepsilon_n)_{n \geq 1}$ be a sequence of i.i.d. random variables with

$$\mathbb{P}(\varepsilon_n = +1) = \mathbb{P}(\varepsilon_n = -1) = \frac{1}{2}.$$

Define the natural filtration

$$\mathcal{F}_n = \sigma(\varepsilon_1, \dots, \varepsilon_n).$$

It is useful to think of the sample space as the set of infinite paths in a binary tree, or equivalently as the interval $[0, 1]$ divided dyadically.

Define the process $(M_n)_{n \geq 1}$ by

$$M_n = \begin{cases} 2^n, & \text{if } \varepsilon_1 = \varepsilon_2 = \dots = \varepsilon_n = +1, \\ 0, & \text{otherwise.} \end{cases}$$

- M_n is a martingale.
- $M_\infty = \lim_{n \rightarrow \infty} M_n = 0$ almost surely.
- $\mathbb{E}[M_\infty | \mathcal{F}_n] = 0 \neq M_n$. (not uniformly integrable)

Note: no other definition of M_∞ will fix this.

- Think of M_n as “mass” distributed on sample space/tree as $n \rightarrow \infty$ gets concentrated on the single path $\varepsilon_i = 1 \ \forall i$ which has probability 0 and is therefore “invisible” (think about where in sample space contribution to $\mathbb{E}[M]$ comes from).
- Interpret M_n as a gambler’s wealth betting everything on “heads” each time. Fair game, but loses everything eventually.
- $(-M_n)$ is net wealth of a gambler who can borrow in order to bet, doubling their bet every time they lose (this time $\varepsilon_k = +1$ is a loss). Eventually they win and their net wealth becomes 0. Thus they are guaranteed to **make** money (having initially been -1).
- We can embed this example into continuous time. The coin tosses occur at time $1 - \frac{1}{n}$, $n = 1, 2, \dots$, and $M(t)$ is constant on each interval $[1 - \frac{1}{n}, 1 - \frac{1}{n+1})$, and $M(t) = 0$ for $t \geq 1$. Then M is a **local martingale** (not a continuous one!).

- Let

$$T_n = \inf\{t : M(t) \geq 2^n\} = \begin{cases} 1 - \frac{1}{n} & \text{if } \varepsilon_1 = \dots = \varepsilon_n = +1, \\ \infty & \text{otherwise.} \end{cases}$$

“stop playing after n heads in a row”

-

$$M^{T_n}(1) = \begin{cases} 2^n & \text{if } \varepsilon_1 = \dots = \varepsilon_n = +1, \\ 0 & \text{otherwise.} \end{cases}$$

and

$$\mathbb{E}[M^{T_n}(1)] = 1 \quad \text{— loss of mass doesn't occur.}$$

- M^{T_n} is a true martingale.

We now give an example in which the martingale is continuous. For $t > 0$, define

$$p_t(x) = \frac{1}{\sqrt{2\pi t}} \exp\left(-\frac{x^2}{2t}\right).$$

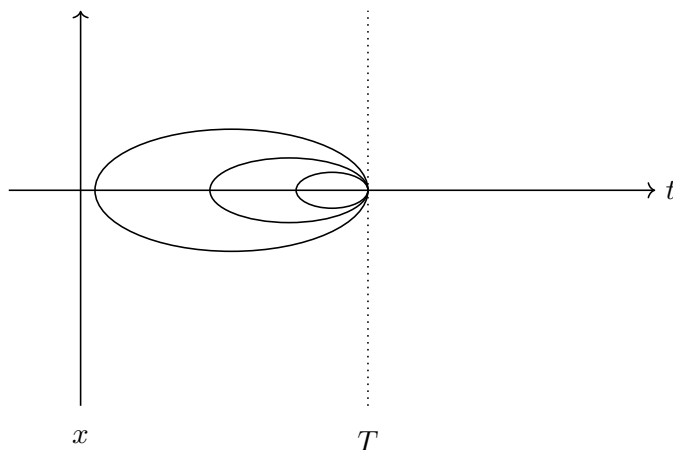
This is the Gaussian density and the transition density of Brownian motion. This solves the **Heat equation**. (Kolmogorov backward equation):

$$\frac{\partial}{\partial t} p_t(x) = \frac{1}{2} \frac{\partial^2}{\partial x^2} p_t(x).$$

Definition of the Process Fix $T > 0$ and define the process $(M_t)_{t \geq 0}$ by

$$M_t = \begin{cases} p_{T-t}(B_t), & t < T, \\ 0, & t \geq T. \end{cases}$$

Contour plot showing singularity of $p_{T-t}(x)$ at $x = 0, t = T$



Itô's Formula is applicable to $p_{T-t}(B_t)$ and $t < T$

$$dM_t = \frac{\partial f}{\partial t}(t, B_t) dt + \frac{\partial f}{\partial x}(t, B_t) dB_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2}(t, B_t) dt.$$

B misses the singularity and there is enough smoothness on the line $\{t = T\} \cap \{x \neq 0\}$ to conclude that M is a local martingale.

Let

$$T_n = \inf\{t \geq 0 : P_{T-t}(B_t) \geq n\}.$$

“Stops B if it gets close to singularity” and M^{T_n} is a true (bounded) martingale.

But

$$\mathbb{E}[M_t] = \begin{cases} P_T(0) & t < T, \\ 0 & t \geq T. \end{cases}$$

So M is not a true martingale — loss of mass at time T . See problem sheet Q3 for example where “mass is lost at all times”.

Some general results:

- A non-negative local martingale is a **supermartingale**

$$\mathbb{E}[M_t^{T_n} | \mathcal{F}_s] = M_s^{T_n} \quad \text{all } s \leq t.$$

(by applying Fatou as $n \rightarrow \infty$)

and so

$$\mathbb{E}[M_s] \geq \mathbb{E}[M_t]$$

Moreover it is a true martingale if and only if $\mathbb{E}[M_t] = \text{constant}$.

- A bounded local martingale is always a true martingale.
- If

$$\mathbb{E} \left[\int_0^t H_s^2 ds \right] < \infty \quad \forall t$$

then

$$\int_0^t H_s dB_s$$

is a true (square integrable) martingale. Which is a special case of M a continuous local martingale and if $M_0 \in L^2$ and $\mathbb{E}[\langle M \rangle_t] < \infty$ then $(M_s; s \leq t)$ is an L^2 -bounded martingale ([1], Corollary 1.25, Chapter IV)

5 Lecture 5 - Girsanov

Standard set up: $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$

Let $\mathbb{Q}$ be another probability measure which is absolutely continuous w.r.t. $\mathbb{P}$ on each $\mathcal{F}_t$, then there exists a non-negative martingale

$(D_t)_{t \geq 0}$ with $\mathbb{E}[D_t] = 1$ s.t.

$$\mathbb{Q}|_{\mathcal{F}_t} = D_t \cdot \mathbb{P}|_{\mathcal{F}_t}.$$

$\mathbb{Q}$ is equivalent of $\mathbb{P}$ on each $\mathcal{F}_t$ if and only if D_t is strictly positive a.s. Meaning $\mathbb{Q}(A) = \mathbb{E}_{\mathbb{P}}[D_t 1_A] \quad \forall A \in \mathcal{F}_t$

5.1 Examples 1:

(1): Let B is a BM under $\mathbb{P}$ (starting from 0) and

$$\mathbb{Q}|_{\mathcal{F}_t} = \exp\left(\lambda B_t - \frac{\lambda^2 t}{2}\right) \cdot \mathbb{P}|_{\mathcal{F}_t}.$$

$\lambda \in \mathbb{R}$ some constant. Let's compute the distribution of B_t under $\mathbb{Q}$:

$$\begin{aligned} \mathbb{E}_{\mathbb{Q}}[f(B_t)] &= \mathbb{E}_{\mathbb{P}}\left[\exp\left(\lambda B_t - \frac{\lambda^2 t}{2}\right) f(B_t)\right] \\ &= \int_{-\infty}^{\infty} f(x) e^{\lambda x - \lambda^2 t/2} \frac{e^{-x^2/2t}}{\sqrt{2\pi t}} dx \end{aligned}$$

where $\frac{e^{-x^2/2t}}{\sqrt{2\pi t}}$ is the density of B_t under $\mathbb{P}$.

$$= \int_{-\infty}^{\infty} f(x) e^{-(x-\lambda t)^2/2t} dx$$

where $\frac{1}{\sqrt{2\pi t}} e^{-(x-\lambda t)^2/2t}$ is the density of B_t under $\mathbb{Q}$.

More generally, B is a **BM with drift** λ under $\mathbb{Q}$

i.e.

$$B_t = \tilde{B}_t + \lambda t \quad \text{where } \tilde{B} \text{ is a } \mathbb{Q}\text{-BM.}$$

Note $\mathbb{P}$ and $\mathbb{Q}$ are not equivalent on $\mathcal{F}_{\infty} = \sigma(\cup_{t \geq 0} \mathcal{F}_t)$

because

$$\frac{B_t}{t} \rightarrow 0 \text{ a.s. } \mathbb{P}, \quad \frac{B_t}{t} \rightarrow \lambda \text{ a.s. } \mathbb{Q},$$

as $t \rightarrow \infty$.

5.2 Examples 2:

Suppose

$$\mathbb{Q}|_{\mathcal{F}_t} = \frac{P_{T-t}(B_t)}{P_T(0)} \cdot \mathbb{P}|_{\mathcal{F}_t}, \quad t < T.$$

Then under $\mathbb{Q}$, B is a **Brownian bridge**, a BM “conditioned to satisfy $B_T = 0$ ”. Which satisfies

$$dB_t = d\tilde{B}_t - \frac{B_t}{T-t} dt, \quad \text{where } \tilde{B} \text{ is a } \mathbb{Q}\text{-BM.}$$

$\mathbb{Q}$ is not absolutely continuous w.r.t. $\mathbb{P}$ on $\mathcal{F}_T$ because

$$\mathbb{Q}(B_T = 0) = 1 \quad \text{but} \quad \mathbb{P}(B_T = 0) = 0.$$

and this corresponds somehow to the loss of mass of the local martingale $P_{T-t}(B_t)$ at $t = T$.

How do martingales behave when we change to an equivalent measure?

Let $\mathbb{Q}|_{\mathcal{F}_t} = D_t \cdot \mathbb{P}|_{\mathcal{F}_t}$. A process $(M_t)_{t \geq 0}$ is a $\mathbb{Q}$ -martingale iff $(D_t M_t)_{t \geq 0}$ is a $\mathbb{P}$ -martingale because

$$\mathbb{E}_{\mathbb{Q}}[M_t | \mathcal{F}_s] = \frac{1}{D_s} \mathbb{E}_{\mathbb{P}}[D_t M_t | \mathcal{F}_s].$$

Assume D is continuous.

5.3 Girsanov's theorem

Every $\mathbb{P}$ -continuous semimartingale is a $\mathbb{Q}$ -semimartingale with the same quadratic variation.

If M is a $\mathbb{P}$ -continuous local martingale then

$$\tilde{M}_t = M_t - \int_0^t \frac{d\langle M, D \rangle_s}{D_s}$$

is a $\mathbb{Q}$ -continuous local martingale. Examples (1) and (2) are both applications of this formula.

5.4 Exponential martingales

If L is a local martingale. then

$$D_t = \mathcal{E}(L)_t = \exp \left(L_t - \frac{1}{2} \langle L \rangle_t \right)$$

is a local martingale too, called the **stochastic exponential** of L . In fact $dD_t = D_t dL_t$ and consequently $M_t - \int_0^t \frac{d\langle M, D \rangle_s}{D_s} = M_T - \langle M, L \rangle_t$

Warning: Given some local martingale (L) satisfying $L_0 = 0$ try to define a $\mathbb{Q}$ from a $\mathbb{P}$ via

$$\mathbb{Q}|_{\mathcal{F}_t} = \mathcal{E}(L)_t \cdot \mathbb{P}|_{\mathcal{F}_t}$$

but we have to check $\mathcal{E}(L)$ is a true martingale i.e. $\mathbb{E}[\mathcal{E}(L)_t] = 1 \quad \forall t$. A famous test for this is Novikov's criterion ([1], chapter VIII Prop 1.15). On the other hand, if you know $\mathbb{Q}$ is a probability measure then $\mathcal{E}(L)_t$ must be a true martingale and Novikov is unnecessary / may not even work.

FINANCIAL MARKETS AND ARBITRAGE

2

6 Lecture 6

6.1 Standard Set Up:

Asset prices $(S_t^0, S_t^1, \dots, S_t^d)$ are assumed to be **non-negative continuous semimartingales** on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t), \mathbb{P})$.

- $\mathbb{P}$: - real world measure
- $(\mathcal{F}_t)_{t \geq 0}$ - market filtration

Often $S_t^0 = \exp\left(\int_0^t r_s ds\right)$ is money account with interest rate, (r_s) . But in general we just need $(S_t^0; t \geq 0)$ to be strictly positive. Then S^0 can be used as a numeraire, i.e. we consider the relative or discounted prices $X_t^i = S_t^i / S_t^0$.

A self-financing strategy is $\phi_t = (\phi_t^0, \phi_t^1, \dots, \phi_t^d)$ predictable, integrable processes so $\Pi_t(\phi) = \sum_0^d \phi_t^i S_t^i$ (value of portfolio consisting of ϕ_t^i units of i th asset at time t) satisfies $d\Pi_t = \sum_0^d \phi_t^i dS_t^i$ i.e. change in portfolio value corresponds to changes in asset prices.

The discounted value of such a portfolio then satisfies

$$d\left(\frac{\Pi_t}{S_t^0}\right) = \sum_{i=1}^d \phi_t^i dX_t^i, \quad (2.1)$$

(note that the sum does not start at 0).

Proof Using Itô's product rule,

$$d\left(\frac{\Pi_t}{S_t^0}\right) = d\Pi_t \left(\frac{1}{S_t^0}\right) + \Pi_t d\left(\frac{1}{S_t^0}\right) + d\Pi_t d\left(\frac{1}{S_t^0}\right).$$

$$d\left(\frac{\Pi_t}{S_t^0}\right) = \sum_i \phi_t^i \frac{dS_t^i}{S_t^0} + \sum_i \phi_t^i S_t^i d\left(\frac{1}{S_t^0}\right) + \sum_i \phi_t^i dS_t^i d\left(\frac{1}{S_t^0}\right).$$

Applying the product rule again,

$$\begin{aligned} d\left(\frac{\Pi_t}{S_t^0}\right) &= \sum_i \phi_t^i \left\{ \frac{dS_t^i}{S_t^0} + S_t^i d\left(\frac{1}{S_t^0}\right) + dS_t^i d\left(\frac{1}{S_t^0}\right) \right\} \\ &= \sum_i \phi_t^i d\left(\frac{S_t^i}{S_t^0}\right). \end{aligned}$$

If $i = 0$, then $d(S_t^0/S_t^0) \equiv 0$.

Conversely, if $\phi_t^1, \dots, \phi_t^d$ and Π_t satisfy (2.1) and we define ϕ_t^0 such that

$$\Pi_t = \phi_t^0 S_t^0 + \sum_{i=1}^d \phi_t^i S_t^i,$$

then $(\phi^0, \dots, \phi^d)$ is self-financing.

A self-financing strategy ϕ is an *arbitrage* if

$$\Pi_0(\phi) \leq 0,$$

and for some deterministic time T ,

$$\Pi_T(\phi) \geq 0 \quad \mathbb{P}\text{-a.s.}, \quad \mathbb{P}(\Pi_T(\phi) > 0) > 0.$$

We also require ϕ to be *admissible*, meaning

$$\frac{\Pi_t(\phi)}{S_t^0}$$

is bounded below by some constant on $[0, T]$ (no unlimited borrowing allowed). Otherwise, an “arbitrage” could be constructed using a local martingale that gains mass.

A probability measure $\mathbb{Q}$ is an equivalent (local) martingale measure (ELMM) if $\mathbb{Q}$ is equivalent to $\mathbb{P}$ on each $\mathcal{F}_t$ and

$$X_t^k = \frac{S_t^k}{S_t^0}$$

is a (local) martingale under $\mathbb{Q}$ for each k .

6.2 Fundamental Theorem of Asset Pricing

$$\exists \text{ ELMM} \Rightarrow \text{NA},$$

and conversely if NA is replaced by NFLVR.

Proof Suppose ϕ is an arbitrage and $\mathbb{Q}$ is an ELMM. Then

$$V_t := \frac{\Pi_t(\phi)}{S_t^0}$$

is a local martingale under $\mathbb{Q}$, since $V_t - V_0$ is a stochastic integral with respect to the discounted asset prices X^k .

If V is bounded below, then by Fatou’s lemma it is a supermartingale, hence

$$0 < \mathbb{E}_{\mathbb{Q}}[V_T] \leq \mathbb{E}_{\mathbb{Q}}[V_0] \leq 0.$$

The strict inequality follows because $\mathbb{P}$ and $\mathbb{Q}$ are equivalent on $\mathcal{F}_T$

6.3 Example: Black-Scholes Model

Let

$$S_t^0 = e^{rt},$$

with constant interest rate r . Let the risky asset satisfy

$$dS_t = S_t(\mu dt + \sigma dB_t),$$

where B is a $\mathbb{P}$ -Brownian motion and μ, σ are constants.

Define $\mathbb{Q}$ by

$$\mathbb{Q}|_{\mathcal{F}_t} = \mathcal{E}(-\alpha B_t) \cdot \mathbb{P}|_{\mathcal{F}_t}, \quad \alpha = \frac{\mu - r}{\sigma}.$$

By Girsanov's theorem,

$$dS_t = S_t(r dt + \sigma d\tilde{B}_t), \quad \tilde{B}_t = B_t + \alpha t,$$

where $\tilde{B}$ is a $\mathbb{Q}$ -Brownian motion.

Solving the SDE,

$$S_t = S_0 \exp\left(\sigma B_t + \left(\mu - \frac{1}{2}\sigma^2\right)t\right).$$

Under $\mathbb{Q}$,

$$S_t \stackrel{(d)}{=} e^X, \quad X \sim N\left(\log S_0 + \left(r - \frac{1}{2}\sigma^2\right)t, \sigma^2 t\right),$$

and

$$\mathbb{E}_{\mathbb{Q}}[e^{-rt}S_t] = S_0.$$

6.4 Bessel Process Example

Let $S_t^0 \equiv 1$ and

$$X_t = x + B_t + \int_0^t \frac{ds}{X_s}.$$

Assume $\mathbb{Q}$ is an ELMM with density

$$D_t = \frac{x}{X_t}.$$

Then

$$dD_t = -\frac{x}{X_t^2} dB_t,$$

so D is a local martingale but not a true martingale. Hence no ELMM exists.

To construct an arbitrage, let

$$V_t = u(t, X_t).$$

By Itô's formula,

$$dV_t = \frac{\partial u}{\partial x}(t, X_t) dX_t + \left(\frac{\partial u}{\partial t} + \frac{1}{2} \frac{\partial^2 u}{\partial x^2}\right)(t, X_t) dt.$$

The strategy is self-financing if u solves the heat equation.

Define

$$u(t, x) = \mathbb{P}(B_T \geq 2x \mid B_t = x) - \mathbb{P}(B_T \leq 0 \mid B_t = x).$$

The portfolio holdings are

$$\phi_t^{(1)} = \frac{\partial u}{\partial x}(t, X_t), \quad \phi_t^{(0)} = u(t, X_t) - \phi_t^{(1)} X_t.$$

7 Lecture 7: Pricing and Replication

We go back to our general set-up for a financial market.

A claim $C \geq 0$ settled at time T can be replicated if there exists a self-financing (and admissible) strategy ϕ such that

1. $\Pi_T(\phi) = C$ $\mathbb{P}$ -a.s.,
2. $\Pi_t(\phi)/S_t^0$, $t \leq T$, is a true martingale under some ELMM $\mathbb{Q}$.

We then say the fair value for C at time $t < T$ is $\pi_t(C)$ which satisfies

$$\pi_t(C) = \Pi_t(\phi) = S_t^0 \mathbb{E}_{\mathbb{Q}} \left[\frac{C}{S_T^0} \mid \mathcal{F}_t \right],$$

because of (ii).

Note

(a) Suppose $\tilde{\phi}$ is another self-financing and admissible strategy with $\Pi_T(\tilde{\phi}) = C$. Recall that

$$d \left(\frac{\Pi_t(\tilde{\phi})}{S_t^0} \right) = \sum_{i=1}^d \tilde{\phi}_t^i dX_t^i,$$

where discounted asset prices are local martingales under $\mathbb{Q}$. Hence $\Pi_t(\tilde{\phi})/S_t^0$ is itself a local martingale under $\mathbb{Q}$.

Since the strategy is admissible, it is a supermartingale by Fatou's lemma, and

$$\Pi_0(\tilde{\phi}) \geq S_0^0 \mathbb{E}_{\mathbb{Q}} \left[\frac{C}{S_T^0} \right] = \Pi_0(\phi).$$

Thus the fair value represents the least cost of setting up a portfolio which hedges C , i.e. an admissible self-financing strategy worth C at time T .

(b) $\Pi_t(C)$ is sometimes called a no-arbitrage price for C . If we add an extra asset to the financial market trading at price $\Pi_t(C)$, then the enlarged market will still be arbitrage-free because $\mathbb{Q}$ remains an ELMM.

(c) One cannot, in general, simply define the fair value by

$$S_t^0 \mathbb{E}_{\mathbb{Q}} \left[\frac{C}{S_T^0} \mid \mathcal{F}_t \right].$$

If a replicating strategy does not exist, the ELMM $\mathbb{Q}$ is not unique and this value depends on the choice of $\mathbb{Q}$. This is the typical situation in incomplete markets.

A spectacular example of a poor strategy ϕ corresponds to a local martingale $P_{T-t}(B_t)$ which hedges the claim $C \equiv 0$ at time T but has positive initial cost.

7.1 Black-Scholes

We consider one risky asset following a geometric Brownian motion. Let $C = f(S_T)$ be a European claim settled at time T , for example

$$f(S_T) = \begin{cases} (S_T - K)^+ & \text{call option,} \\ (K - S_T)^+ & \text{put option,} \\ \mathbf{1}_{\{S_T \geq K\}} & \text{binary option.} \end{cases}$$

If the claim is replicable, its fair price at time $t \leq T$ is

$$\Pi_t(C) = e^{rt} \mathbb{E}_{\mathbb{Q}}[e^{-rT} f(S_T) \mid \mathcal{F}_t].$$

By the Markov property, $\Pi_t(C) = u(t, S_t)$, where $\mathbb{Q}$ is the equivalent martingale measure. In the case of a call option, u is given by the Black-Scholes formula.

7.2 Replication argument

Define u by the conditional expectation above and note by the tower property that

$$V_t := e^{-rt} \Pi_t(C) = e^{-rt} u(t, S_t)$$

is a martingale.

Assume u is smooth. Applying Itô's formula,

$$\begin{aligned} V_t - V_0 &= \int_0^t e^{-rs} \frac{\partial u}{\partial x}(s, S_s) dS_s + \int_0^t e^{-rs} \left(\frac{1}{2} \sigma^2 S_s^2 \frac{\partial^2 u}{\partial x^2}(s, S_s) + \frac{\partial u}{\partial t}(s, S_s) - ru(s, S_s) \right) ds \\ &= \int_0^t \frac{\partial u}{\partial x}(s, S_s) dX_s, \end{aligned}$$

where $X_t = e^{-rt} S_t$ is a $\mathbb{Q}$ -martingale and

$$dX_t = e^{-rt} dS_t - re^{-rt} S_t dt.$$

The vanishing of the drift term implies that u solves the Black-Scholes PDE:

$$\frac{1}{2} \sigma^2 x^2 \frac{\partial^2 u}{\partial x^2} + rx \frac{\partial u}{\partial x} - ru + \frac{\partial u}{\partial t} = 0.$$

If ϕ is self-financing, then

$$d(e^{-rt} \Pi_t(\phi)) = \phi_t^{(1)} dX_t.$$

Thus we choose

$$\phi_t^{(1)} = \frac{\partial u}{\partial x}(t, S_t),$$

the Black-Scholes delta, and

$$e^{rt} \phi_t^{(0)} = u(t, S_t) - \phi_t^{(1)} S_t.$$

This gives a self-financing strategy that replicates the claim.

7.3 Predictable Representation Property (PRP)

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space. A continuous local martingale X has the predictable representation property (PRP) if every local martingale M can be written as

$$M_t = M_0 + \int_0^t H_s dX_s$$

for some predictable process H .

Theorem If $(\mathcal{F}_t)$ is the augmented filtration generated by a Brownian motion B , then B has the PRP.

7.4 Application to Black–Scholes

Assume the market filtration is generated by the risky asset. Then it is also generated by the $\mathbb{Q}$ -Brownian motion $\tilde{B}$. For any integrable claim C , define the $\mathbb{Q}$ -martingale

$$V_t = \mathbb{E}_{\mathbb{Q}}[e^{-rT}C \mid \mathcal{F}_t] = V_0 + \int_0^t H_s d\tilde{B}_s.$$

Since $dX_s = \sigma X_s d\tilde{B}_s$, this can be written as

$$V_t = V_0 + \int_0^t \frac{H_s}{\sigma X_s} dX_s.$$

Hence the replicating strategy is given by

$$\phi_t^{(1)} = \frac{H_t}{\sigma X_t}.$$

The Black–Scholes market is therefore complete: every integrable claim can be replicated.

If the market is complete, the ELMM is unique. Let $A \in \mathcal{F}_T$ and consider the claim $C = \mathbf{1}_A S_T^0$. Suppose it is replicated by a self-financing strategy ϕ such that $\Pi_t(\phi)/S_t^0$ is a martingale under an ELMM $\mathbb{Q}$.

Now let $\tilde{\mathbb{Q}}$ be another ELMM. Since $\Pi_t(\phi)/S_t^0$ is a stochastic integral with respect to discounted asset prices, it is a local martingale under $\tilde{\mathbb{Q}}$. Being bounded, it is a true martingale.

Assuming $\mathcal{F}_0$ is trivial,

$$\begin{aligned} \tilde{\mathbb{Q}}(A) &= \mathbb{E}_{\tilde{\mathbb{Q}}}[\mathbf{1}_A] = \frac{1}{S_0^0} \mathbb{E}_{\tilde{\mathbb{Q}}} \left[\frac{C}{S_T^0} \right] \\ &= \frac{1}{S_0^0} \mathbb{E}_{\tilde{\mathbb{Q}}} \left[\frac{\Pi_T(\phi)}{S_T^0} \right] = \frac{\Pi_0(\phi)}{S_0^0} = \mathbb{Q}(A). \end{aligned}$$

Thus $\tilde{\mathbb{Q}} = \mathbb{Q}$, and the ELMM is unique.

8 Lecture 8: Implied Volatility

8.1 Black-Scholes Formula for a Call Option

The value of a European call option under the Black-Scholes model is

$$C_{\text{BS}} = S_0 \Phi(d_+) - K e^{-rT} \Phi(d_-),$$

where

$$d_{\pm} = \frac{\ln(S_0/K) + \left(r \pm \frac{1}{2}\sigma^2\right)T}{\sigma\sqrt{T}}.$$

8.2 Behavior of $d_{\pm}$ and C_{BS}

- As $\sigma \rightarrow \infty$:

$$d_{\pm} \rightarrow \pm\infty \implies C_{\text{BS}} \rightarrow S_0.$$

- As $\sigma \rightarrow 0$:

$$d_{\pm} \rightarrow \begin{cases} +\infty & S_0 > K e^{-rT} \\ -\infty & S_0 < K e^{-rT} \end{cases} \implies C_{\text{BS}} \rightarrow (S_0 - K e^{-rT})^+.$$

- Moreover,

$$\frac{\partial C_{\text{BS}}}{\partial \sigma} > 0 \implies \sigma \mapsto C_{\text{BS}}(\sigma) \text{ is strictly increasing.}$$

8.3 Definition: Implied Volatility

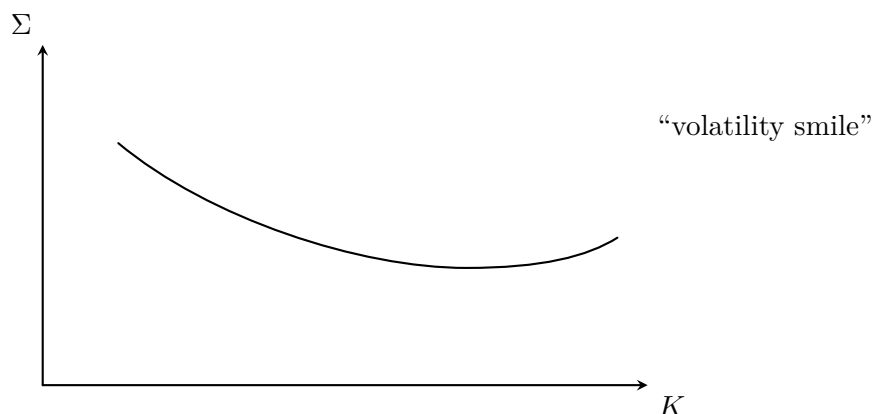
Given a market call price

$$C \in ((S_0 - K e^{-rT})^+, S_0),$$

there exists a unique Σ such that $C = BS(\Sigma)$, with all other parameters fixed. This Σ is called the *implied volatility*.

8.4 Why Implied Volatility Matters

European call options are traded on exchanges, and their prices are determined by supply and demand — i.e., market sentiment. The volatility in the Black-Scholes model refers to the future; in stable conditions, implied volatilities may be similar to historical volatility observed from past prices of the underlying. However, when implied volatilities are calculated for different strikes and maturities, we typically observe patterns such as the *volatility smile or skew*:



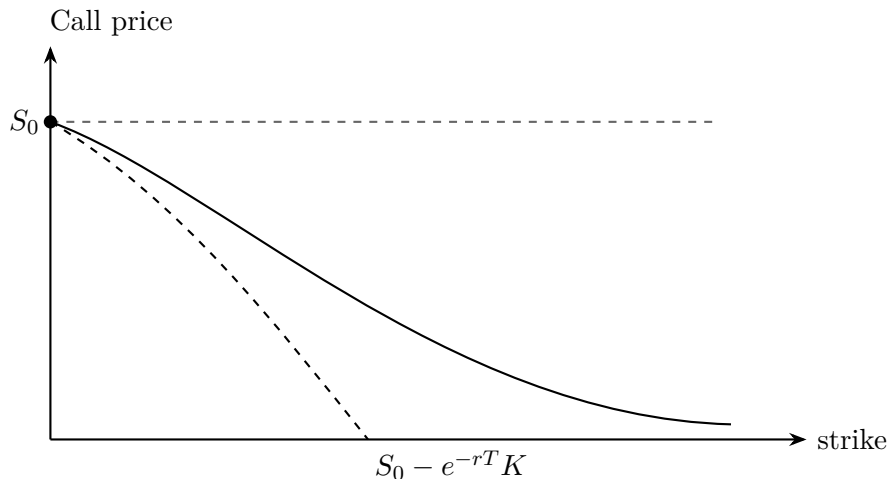
Thus the Black-Scholes model cannot be fitted to the real financial market - if it could Σ would be constant!. What to do about that? Why does this happen?

9 Lecture 9: Volatility Distribution

9.1 Properties of observed call prices

Fix a maturity T , consider call prices $C(K)$ of varying strikes. (Assume a fixed interest rate r .)

(Sketch: call price vs strike: decreasing convex curve with $C(0) = S_0$ and lower bound $S_0 - e^{-rT}K$.)



$K \mapsto C(K)$ satisfies:

- 1) $C(0) = S_0$ (current value).
- 2) $K \mapsto C(K)$ is non-increasing, with $\lim_{K \rightarrow \infty} C(K) = 0$
- 3) $K \mapsto C(K)$ is *convex*.
- 4) $C(K) \geq S_0 - e^{-rT}K \Rightarrow C'_+(0) \geq -e^{-rT}$.

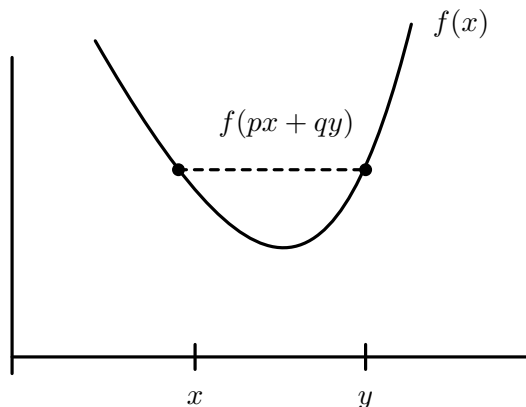
These properties can be explained by arbitrage arguments (see problem sheet 3).

9.2 Convexity and one-sided derivatives

Convex means for all $p, q \in \mathbb{R}$ such that $p + q = 1$, $p, q \geq 0$, $x \leq y$ then

$$f(px + qy) \leq pf(x) + qf(y)$$

Graphically:



Convex doesn't quite mean differentiable with $f''(x) \geq 0 \forall x$. Examples: $f(x) = x^2$, $f(x) = |x|$.

However a convex function always has left + right derivatives

$$f'_{\pm}(a) = \lim_{\substack{h \downarrow 0, h > 0 \\ \text{or } h \uparrow 0, h < 0}} \frac{f(a+h) - f(a)}{h}.$$

which are non-decreasing, and $f'_-(a) \leq f'_+(a)$ always, all left/right continuous respectively.

Moreover we recover f from either of them by integration:

$$f(b) = f(a) + \int_a^b f'_{\pm}(x) dx.$$

Proposition: This is a 1 – 1 correspondence between probability measure μ on $[0, \infty]$ with mean $\int_0^\infty x \mu(dx) = \mu < \infty$ and functions $f : [0, \infty) \rightarrow \mathbb{R}$ which are convex, non-increasing with $f(-\infty) = 0$ and $f'_+(0) \geq -1$.

This is given by

$$f(\theta) = \int (x - \theta)^+ \mu(dx), \quad \text{and} \quad \mu(\theta, \infty) = 1 - \mu[0, \theta] = -f'_+(\theta).$$

Note: if f is twice differentiable then μ has a density given by $f''_+(\theta)$ except possibly at origin.

Note: if $f'_+(0) > -1$ then $\mu(\{0\}) > 0$

Sketch proof. Start with f , then $\theta \mapsto 1 + f'_+(\theta)$ has the properties of a c.d.f of some measure μ on $[0, \infty)$.

Moreover

$$\begin{aligned} f(\infty) - f(\theta) &= \int_{\theta}^{\infty} f'_+(y) dy = - \int_{\theta}^{\infty} \mu(y, \infty) dy \\ &= - \int_{\theta}^{\infty} \int_0^{\infty} \mathbf{1}(x > y \geq \theta) \mu(dx) dy \\ (\text{Fubini}) &= - \int_0^{\infty} (x - \theta)^+ \mu(dx) \end{aligned}$$

.

Conversely, can start with μ , define f , and then differentiate:

$$\lim_{h \downarrow 0} \frac{f(\theta+h) - f(\theta)}{h} = \lim_{h \downarrow 0} \frac{\int (x - \theta - h)^+ - (x - \theta)^+ \mu(dx)}{h}.$$

9.3 Link to risk-neutral distribution

We apply this to the call option prices we observe, taking $f(K) = e^{rt}C(K)$, and obtain a probability measure we call the implied distribution of the underlying at time T . This is because if the observed prices were coming from some model $\mathbb{Q}$, then we would have

$$e^{rT}C(K) = \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+],$$

and so μ would be the distribution of S_T under $\mathbb{Q}$.

Important consequence: From the observed call prices We can consistently price any European option with payoff $h(S_T)$: just compute the expectation of this payoff w.r.t. the implied distribution of S_T .

In fact we can replicate $h(S_T)$ using a portfolio of puts + calls with (a continuum of) different strikes.

See P.S. Q2. (Carr–Madan formula)

10 Lecture 10: Local Volatility models

The asset price evolves under the ELMM $\mathbb{Q}$ according to

$$dS_t = S_t(r dt + \sigma(t, S_t) dB_t),$$

where B is a $\mathbb{Q}$ -BM and $\sigma(\cdot, \cdot)$ is some deterministic function called the *local volatility*. We assume existence and uniqueness for this SDE; S is a time-inhomogeneous diffusion.

example (1) $\sigma(t, x) = \sigma(t)$, a function of time alone.

example (2) CEV model:

$$dS_t = \sigma S_t^\beta dB_t + r S_t dt.$$

- $\beta = 1$ is Black–Scholes.
- $\beta \in (0, 1)$ then local vol $\sigma S_t^{\beta-1}$ is a *decreasing* function of price. For β in this range S hits zero which is an absorbing point (bankruptcy).

example (1) continued... $\sigma(t, x) = \sigma(t)$, So $dS_t = S_t(r dt + \sigma(t) dB_t)$.

Solution:

$$S_t = S_0 \exp\left(\int_0^t \sigma(s) dB_s + rt - \frac{1}{2} \int_0^t \sigma(s)^2 ds\right).$$

Check by Itô. Because $\sigma(\cdot)$ is deterministic,

$$\int_0^t \sigma(s) dB_s \sim N\left(0, \int_0^t \sigma(s)^2 ds\right),$$

and S_t has a lognormal distribution.

In fact, for a fixed T ,

$$S_T \stackrel{d}{=} \tilde{S}_T,$$

where $\tilde{S}$ solves the GBM SDE

$$d\tilde{S}_t = \tilde{S}_t(r dt + \Sigma dB_t),$$

if we choose the constant Σ so that

$$\Sigma^2 T = \int_0^T \sigma(t)^2 dt.$$

Hence the prices of call options expiring at T in the model agree with those in Black–Scholes with volatility Σ , Σ is the implied volatility.

Also Σ doesn't depend on the strike – so flat not a smile.

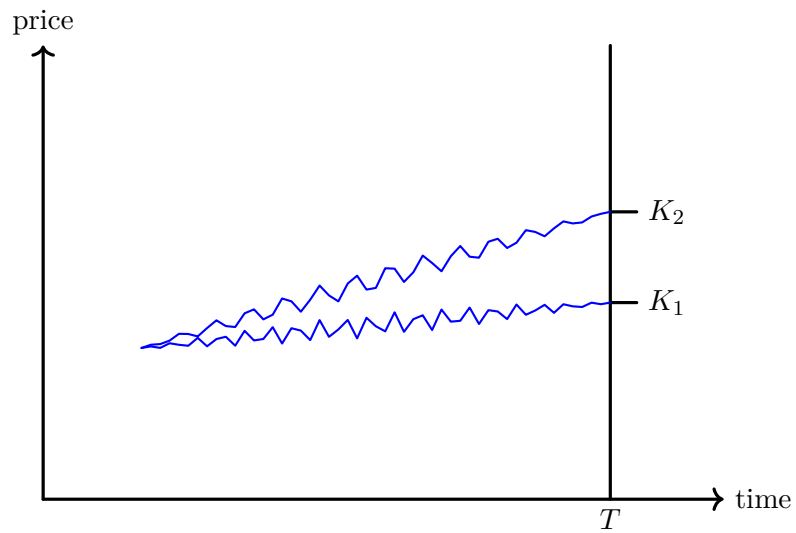
- If $\beta > 1$ then the local volatility is an increasing function of the price. S remains +ve for all time but $e^{-rt}S_t$ is a strict local martingale – which is interpreted as a bubble (see PS4). Model for cryptocurrency?
- The market is complete (because of PRP of BM) and the ELMM unique.
- How does implied vol behave? Investigate numerically.

$\beta = 1$, constant because BS.

$\beta \in (0, 1)$: implied vol is decreasing as the strike increases.

$\beta > 1$: implied vol is increasing as the strike increases.

Heuristic reason: A path of S contributing to the value of option (ie ending above strike) tends to be at higher level, and therefore experience higher local volatilities (if $\beta > 1$), when the strike increases.



How to fit a CEV model (or any other parameterised local vol model): compare call prices with different strikes and expiry times computed from the model with observed prices for those calls. Or equivalently compare implied vols computed from model prices with implied vols observed in the market.

11 Lecture 11: Dupire's Formula

Question: If we can use any local vol $\sigma(\cdot, \cdot)$ (rather than just a function from a parametric family) can we fit the currently observed call prices perfectly?

Since the observed calls determine implied distributions the question becomes, choose $\sigma(\cdot, \cdot)$ such that distribution S_t given by those implied distributions for all $t \geq 0$?

Let $y \mapsto p_t(x, y)$ be the density of the distribution of S_t (we fix $S_0 = x$ once and for all) under the local volatility model with some $\sigma(\cdot, \cdot)$. We want to find σ so that

$$p_t(x, y) = e^{rt} C''(y, t) \quad \forall t, y$$

where C is the observed price of call with strike y and time t .

From now on take C to be smooth. Now $p_t(\cdot, \cdot)$ solves the Kolmogorov equation, but because x is fixed and y varies we need the *forward equation*.

11.1 Interlude on the forward equation

Consider a general SDE for a diffusion

$$X_t = x + \int_0^t \sigma(s, X_s) dB_s + \int_0^t b(s, X_s) ds$$

with $(\sigma(s, x))^2 = a(s, x)$.

Take a smooth and compactly supported test function h and apply Itô to $h(X_t)$:

$$h(X_t) = h(X_0) + \int_0^t h'(X_s) \sigma(s, X_s) dB_s + \int_0^t (Lh)(s, X_s) ds$$

where

$$Lh(s, x) = b(s, x) \frac{\partial h}{\partial x} + \frac{1}{2} a(s, x) \frac{\partial^2 h}{\partial x^2}$$

is the generator of X .

Take expectations because h is compactly supported the stochastic integral is a true martingale.

Now differentiate wrt t :

$$\frac{d}{dt} \mathbb{E}[h(X_t)] = \mathbb{E}[Lh(t, X_t)].$$

re-write with density p

$$\int h(y) \frac{\partial p_t(x, y)}{\partial t} dy = \int (Lh(t, y)) p_t(x, y) dy.$$

Now can do integration by parts to shift the derivatives on h onto p on the R.H.S.

we get

$$\int h(y) L^* p_t(x, y) dy.$$

where

$$L^* p_t = -\frac{\partial}{\partial y} (b(s, y) p_t(x, y)) + \frac{1}{2} \frac{\partial^2}{\partial y^2} (a(s, y) p_t(x, y)).$$

Since h is arbitrary we must have

$$\boxed{\frac{\partial}{\partial t} p_t(x, y) = L^* p_t(x, y)} \quad \leftarrow \text{Forward equation}$$

Plug into forward equation our desired relationship between p and C .

Note $b(t, y) = ry$

$$a(t, y) = y^2 \sigma^2(t, y) \quad \leftarrow \quad \text{care}$$

$$\frac{\partial}{\partial t} \left(e^{rt} \frac{\partial^2}{\partial y^2} C(t, y) \right) = - \frac{\partial}{\partial y} \left(r e^{rt} y \frac{\partial^2}{\partial y^2} C(t, y) \right) + \frac{1}{2} \frac{\partial^2}{\partial y^2} \left(y^2 \sigma^2(t, y) \frac{\partial^2}{\partial y^2} C(t, y) \right).$$

Re-arrange, cancel e^{rt} and simplify derivatives,

$$\begin{aligned} \frac{1}{2} \frac{\partial^2}{\partial y^2} \left(y^2 \sigma^2(t, y) \frac{\partial^2}{\partial y^2} C(t, y) \right) &= \frac{\partial^3}{\partial t \partial y^2} C(t, y) + 2r \frac{\partial^2}{\partial y^2} C(t, y) + ry \frac{\partial^3}{\partial y^3} C(t, y) \\ &= \frac{\partial^2}{\partial y^2} \left(\frac{\partial}{\partial t} C(t, y) + ry \frac{\partial}{\partial y} C(t, y) \right). \end{aligned}$$

Thus integrating twice (and setting the constants of integration to 0) gives *Dupire's relation*:

$$\sigma(t, y)^2 = \frac{2 \left(\dot{C}(t, y) + ry C'(t, y) \right)}{y^2 C''(t, y)}.$$

Note:

1. numerator is +ve because calls satisfy

$$t \mapsto C(t, e^{rt} y) \text{ is increasing for each } y;$$

otherwise there would be calendar arbitrages (see PS 3 Q1 for $r = 0$ case).

2. We should check this σ does correspond to a well defined process - but we won't!
3. This local vol fits the (currently) implied one-dimensional marginals of S_t perfectly, but it may not describe the dynamics of S correctly. Manifest in
 - As time evolves will not stay calibrated to calls correctly.
 - It could (will) price options that depend on multi-dimensional dists of S incorrectly.

12 Lecture 12: Stochastic volatility models

The most general form of a stochastic volatility model is

$$dS_t = S_t(r dt + \sigma_t dB_t)$$

with B a BM under an ELMM $\mathbb{Q}$, r the fixed interest rate, and σ_t some *random process*.

In this general sense local volatility models are also stochastic volatility models, but more typically in a stochastic volatility model a second SDE is given describing the behaviour of σ_t :

$$d\sigma_t = \alpha(\sigma_t, S_t) dt + \beta(\sigma_t, S_t) dW_t$$

where α, β given functions and W another $\mathbb{Q}$ -BM.

12.1 Examples: Heston's model

$$dS_t = rS_t dt + \sqrt{v_t} S_t dB_t$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t$$

Here dv_t is the *instantaneous variance*, $\kappa(\theta - v_t)$ is the *mean reversion term*, and $\xi \sqrt{v_t}$ is the “*volatility of variance*”.

$$dB_t dW_t = \rho dt \quad (\text{where } \rho \text{ is the correlation})$$

- if $\xi = 0$, then $v_t \equiv \theta \forall t$ solves 2nd SDE and we reduce to Black-Scholes.
- (S, v) is a two dimensional diffusion, so Markov, but we do not expect S alone to be Markov
- Take $\kappa, \theta \geq 0$ so θ is long term average for v .
- $v_t \geq 0 \forall t$ and if $2\kappa\theta > \xi^2$ is strictly +ve all times.
- if $\rho \in (-1, +1)$ the ELMM is not unique. In particular under other EMMs the values of κ, θ can be different. v is not a tradable asset.

12.2 Stochastic alpha beta rho (SABR) model

$$dS_t = \alpha_t S_t^\beta dB_t + rS_t dt$$

$$d\alpha_t = v\alpha_t dW_t \quad dB_t dW_t = \rho dt$$

- Natural extension of the CEV model in which the volatility parameter α evolves according to a geometric BM.
- It exhibits bubbles and bankruptcy like CEV.
- Once again if $\rho \in (-1, +1)$ it is incomplete and ELMM is not unique.

12.3 More about incompleteness

Suppose B and W are two BMs (wrt to same prob measure and filtration) with $dB dW = \rho dt$ and $\rho \in (-1, +1)$.

Define $B_t^\perp = \frac{1}{\sqrt{1-\rho^2}}(W_t - \rho B_t)$ so that $W_t = \rho B_t + \sqrt{1-\rho^2} B_t^\perp$

We check that $B^\perp$ is a BM with $dB dB^\perp = 0$ (so B is independent of $B^\perp$). Now suppose we have a stochastic volatility model of the form considered previously with B, W the $\mathbb{Q}$ -BM driving the SDEs for S and σ respectively.

Define $\tilde{\mathbb{Q}}$ via

$$\left. \frac{d\tilde{\mathbb{Q}}}{d\mathbb{Q}} \right|_{\mathcal{F}_t} = \exp \left(\int_0^t \theta_s dB_s^\perp - \frac{1}{2} \int_0^t \theta_s^2 ds \right)$$

where θ is some process such that the exponential is a true martingale (eg bounded or even just a constant).

According to Girsanov's theorem:

$$B_t - \langle B, L \rangle_t \text{ is a } \tilde{\mathbb{Q}}\text{-BM}$$

where

$$L_t = \int_0^t \theta_s dB_s^\perp \quad dB dL = \theta dB dB^\perp = 0$$

so B is a $\tilde{\mathbb{Q}}$ -BM itself; it then follows $e^{-rt}S_t$ is a $\tilde{\mathbb{Q}}$ local martingale and $\tilde{\mathbb{Q}}$ is an ELMM.

- Consequently the market is incomplete – it is not possible to hedge claims (including call options typically) by investing in the underlying + money account alone.
- However recall we don't want to hedge exchange traded options such as calls. Instead we usually want to use such options to price + maybe hedge more exotic options (“over the counter”).
- When call options are included as assets that we can trade in, the resulting enlarged model typically becomes complete. Correspondingly the ELMM will typically be uniquely determined by fitting the model to the call prices.

13 Lecture 13: Interest Rate

Financial markets do not have a single interest rate. Instead, they have bonds of various maturities.

A **zero-coupon bond** pays 1 at its maturity T and has price at time $t < T$, $P(t, T) < 1$. If held to maturity the yield (i.e effective interest rate) is defined as:

$$y(t, T) = -\frac{1}{T-t} \log P(t, T) \quad \text{so} \quad P(t, T) e^{y(t, T)(T-t)} = 1$$

The **yield curve**(at an instant in time t) is the mapping $\tau \mapsto y(t, \tau)$.

- By combining bonds with maturities S and T , we can, at a time $t < s < T$ guarantee an interest rate over a future period $[S, T]$ (equivalent to a **forward rate agreement**)
- At time t : Buy 1 T-bond but short $\frac{P(t, T)}{P(t, S)}$ units of S -bond. Net cost is then zero.
- At time s , pay $\frac{P(t, T)}{P(t, S)}$ to cover the liability from S bond.
- At time T , receive 1 from the T-bond.
- Over $[s, T]$ effective interest rate is $-\frac{1}{T-S} \log \frac{P(t, T)}{P(t, S)}$
- The instantaneous forward rate is the limit as $S \uparrow T$: $f(t, T) = -\frac{\partial}{\partial T} \log P(t, T)$

We can recover bond prices from the forward rates by integrating and using $P(T, T) = 1$:

$$P(t, T) = \exp \left\{ - \int_t^T f(t, u) du \right\}$$

Finally we have the short rate,

$$r_t = \lim_{T \downarrow t} f(t, T) = \lim_{T \downarrow t} y(t, T)$$

Note: We are assuming $T \mapsto P(t, T)$ is differential. In fact the yield and forward curves are often smooth function. This is very different from the behavior as t varies, which appears "Brownian". (See NYT chart)

14 Lecture 14: HJM framework

- Traded assets are bonds of all maturities $T > t$
- Additionally there is a money account to which the short rate r_t is applied.

HJM starts with SDEs for the forward rates:

suppose under the real world measure $\mathbb{P}$, for each T , $t \mapsto f(t, T)$ is a semimartingale for $t \in [0, T]$ with

$$d_t f(t, T) = \alpha(t, T) dt + \sigma(t, T) dB_t$$

where α, σ may be random processes adapted to the market filtration $(\mathcal{F}_t)_{t \geq 0}$ for each T , and B is a $(\mathbb{P}, \mathcal{F}_t)$ -BM, same for all T . Can generalise to models with multiple BMs.

Proposition The corresponding evolution of the bond prices is given by

$$d_t P(t, T) = P(t, T) \left\{ \sigma^*(t, T) dB_t + \left(r_t + \alpha^*(t, T) + \frac{1}{2} (\sigma^*(t, T))^2 \right) dt \right\},$$

where

$$\alpha^*(t, T) = - \int_t^T \alpha(t, u) du, \quad \sigma^*(t, T) = - \int_t^T \sigma(t, u) du.$$

Sketch proof Recall

$$P(t, T) = \exp \left(- \int_t^T f(t, u) du \right).$$

So start with

$$\begin{aligned} d_t \left(- \int_t^T f(t, u) du \right) &= f(t, t) dt - \int_t^T d_t f(t, u) du \\ &= r_t dt - \int_t^T \left(\alpha(t, u) dt + \sigma(t, u) dB \right) du \\ &= r_t dt + \alpha^*(t, T) dt + \underbrace{\sigma^*(t, T)}_{\text{("stochastic Fubini")}} dB \end{aligned}$$

Now we just use Itô on $P(t, T)$, defined via the exponential.

14.1 When is the market arbitrage free

We have one source of noise but infinitely many bonds. Unless the bond prices are "consistent" there will be arbitrages (since just one asset (plus the money account) would typically give a complete market, we expect to be able to replicate + price bonds of all maturities by trading in just one).

Can we find an ELMM? We take the money account to which the short rate is applied as numeraire, and so we want the discounted bonds $\exp \left(- \int_0^t r_s ds \right) P(t, T)$ to be local martingales. Consider a change of measure of the form

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \exp \left(- \int_0^t \Theta(s) dB_s - \frac{1}{2} \int_0^t \Theta(s)^2 ds \right),$$

for some Θ such that density is a true martingale. Then

$$B_t = B_t^{\mathbb{Q}} - \int_0^t \Theta(s) ds, \quad \text{where } B_t^{\mathbb{Q}} \text{ is a } \mathbb{Q}\text{-BM.}$$

Substituting into the expression from the proposition we obtain

$$d_t P(t, T) = P(t, T) \left\{ \sigma^*(t, T) dB_t^{\mathbb{Q}} + \left(r_t + \alpha^*(t, T) + \frac{1}{2} (\sigma^*(t, T))^2 - \sigma^*(t, T) \Theta(t) \right) dt \right\},$$

and so the discounted bonds are local martingales if

$$\alpha^*(t, T) + \frac{1}{2} (\sigma^*(t, T))^2 = \sigma^*(t, T) \Theta(t). \quad (*)$$

Equivalently, differentiating in T (very careful about minus signs)

$$\alpha(t, T) = \sigma(t, T) \left(\int_t^T \sigma(t, u) du + \Theta(t) \right) \quad (2.2)$$

This is known as the **HJM drift condition** and we have proved:

Proposition The bond market is arbitrage free if there exists a market price of risk Θ satisfying (2.2) and such that

$$\mathcal{E} \left(- \int_0^t \Theta(s) dB_s \right)$$

is a true martingale.

Note: (2.2) has to hold for all T but Θ only depends on t , so typically cannot solve for arbitrary α and σ .

14.2 Examples

example: $\sigma(t, T) \equiv 0$; no volatility.

So the HJM drift condition implies $\alpha(t, T) \equiv 0$ too.

So each $t \mapsto f(t, T)$ is constant, but the initial curve $T \mapsto f(0, T)$ need not be trivial. So there is some deterministic function, call it $r(\cdot)$ such that

$$f(t, T) = r(T) \quad \text{for all } T \geq t.$$

r is indeed the short rate $r(t) = f(t, t)$. This is the case of a deterministic but time varying interest rate. Notice that “ r reads along the intial forward curve”.

Bond prices are given by

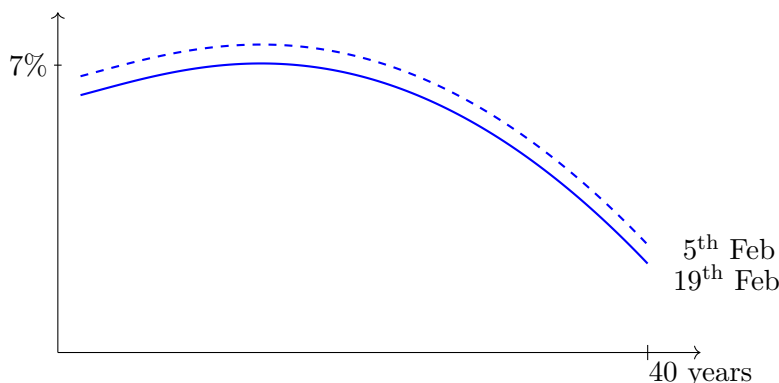
$$P(t, T) = \exp \left(- \int_t^T f(t, u) du \right) = \exp \left(- \int_t^T r(u) du \right).$$

discounted bond prices are given by

$$\exp \left(- \int_0^t r(s) ds \right) P(t, T) = \exp \left(- \int_0^T r(s) ds \right)$$

i.e. non-random constants. The real world measure is (trivially) an EMM

Example: current forward curve from BoE (Bank of England).



– so curve has shifted down slightly.

14.3 Example: Ho-Lee model

$$\sigma(t, T) \equiv \sigma > 0 \quad \text{a constant}$$

Then the HJM drift condition reads

$$\alpha(t, T) = (T - t) \sigma^2 + \Theta(t) \sigma$$

So dynamics of forward curve are:

$$d_t f(t, T) = \left((T - t) \sigma^2 + \Theta(t) \sigma \right) dt + \sigma dB_t.$$

integrate up to obtain (Hint: $\int_0^t (T - s) ds = (T^2 - (T - t)^2)/2 = t(T - t) + t^2/2$)

$$f(t, T) = f(0, T) + \underbrace{\sigma^2 t(T - t) + \frac{\sigma^2 t^2}{2}}_{\text{doesn't depend on } T} + \sigma \int_0^t \Theta(s) ds + \sigma B_t,$$

doesn't depend on T .

Short rate:

$$r(t) = f(t, t) = f(0, t) + \frac{\sigma^2 t^2}{2} + \sigma \int_0^t \Theta(s) ds + \sigma B_t$$

- WARNING: don't attempt to find dr by putting $T = t$ into SDE for $d_t f(t, T)$: wrong answer
- Short rate reads along initial forward curve but with a random perturbation.

If we parameterize the forward curve by $\tau = T - t$, “time ahead”, then,

$$\tau \mapsto f(t, t + \tau) = \underbrace{f(0, t + \tau)}_{\text{left shift of initial curve}} + \underbrace{\frac{\sigma^2 t \tau}{2}}_{\text{linear function of } \tau} + \underbrace{(r(t) - f(0, t))}_{\text{random vertical shift}}$$

Question: can we choose an initial forward curve so the “shape” of forward curve stays the same?
Yes, take

$$f(0, \tau) = -\sigma^2 \frac{\tau^2}{2} + b\tau + c$$

then

$$f(t, t + \tau) = f(0, \tau) + \text{vert shift.}$$

Not a bad first attempt to model what we see in BoE data

15 Lecture 15: Short rate models

15.1 Another Look at Ho-Lee

Bond prices are given by

$$p(t, T) = \exp\left(-\int_t^T f(t, u) du\right)$$

Substitute in the form of $f(t, u)$ from before and we obtain:

$$\begin{aligned} p(t, T) &= \exp\left(-(T-t)r(t) - \sigma^2(T-t)^2/2 - \int_t^T (f(0, u) - f(0, t)) du\right) \\ &= \exp\left(A(t, T)r(t) + B(t, T)\right) \end{aligned}$$

Notice the only randomness is from $r(t)$ because we treat the initial forward curve as known at time zero so deterministic. Any model with bond prices of this form for some functions A and B is called affine. (Because yield is affine function of r .)

Now for a completely different approach to find the same formula: using the EMM. Recall $\mathbb{Q}$ is our ELMM and

$$\underbrace{B_t}_{\mathbb{P}\text{-BM}} = \underbrace{B_t^{\mathbb{Q}}}_{\mathbb{Q}\text{-BM}} - \underbrace{\int_0^t \Theta(s) ds}_{\text{Market price of risk}}$$

We can re-write our previous equation for the short rate as

$$\begin{aligned} r(t) &= f(0, t) + \sigma^2 t^2/2 + \sigma \int_0^t \Theta(s) ds + \sigma B_t \\ &= f(0, t) + \sigma^2 t^2/2 + \sigma B_t^{\mathbb{Q}} \quad \leftarrow \text{Aside: remember } f(0, t) \neq \mathbb{E}_{\mathbb{Q}}^{r(t)} \\ &= r(0) + \int_0^t \mu(s) ds + \sigma B_t^{\mathbb{Q}} \end{aligned}$$

where μ is the infinitesimal drift of r .

Now we assume $\mathbb{Q}$ is a EMM not just an ELMM.

Proposition

$$P(t, T) = \mathbb{E}_{\mathbb{Q}}\left[\exp\left(-\int_t^T r(u) du\right) \mid \mathcal{F}_t\right]$$

Proof. The discounted bond $\exp\left(-\int_0^t r(s) ds\right) P(t, T)$ is a $\mathbb{Q}$ -martingale and $P(T, T) = 1$ so

$$\exp\left(-\int_0^t r(s) ds\right) P(t, T) = \mathbb{E}_{\mathbb{Q}}\left[\exp\left(-\int_0^T r(s) ds\right) \mid \mathcal{F}_t\right] = \exp\left(-\int_0^t r(s) ds\right) \mathbb{E}_{\mathbb{Q}}\left[\exp\left(-\int_t^T r(s) ds\right) \mid \mathcal{F}_t\right].$$

Notice by integrating the expression for r :

$$\int_0^T r(s) ds = T \cdot r(0) + \int_0^T \int_0^s \mu(h) dh ds + \sigma \int_0^T B_s^{\mathbb{Q}} ds$$

So

$$\int_0^T r(s) ds \sim N\left(T r(0) + \int_0^T \int_0^s \mu(h) dh ds, \sigma^2 T^3/3\right)$$

Then by Markov property (careful, it's time inhomogeneous)

$$\int_t^T r(u) du \mid \mathcal{F}_t \sim N\left((T-t)r(t) + \int_t^T \int_t^u \mu(h) dh du, \sigma^2(T-t)^3/3\right)$$

So now use m.g.f. for Gaussian to obtain

$$\begin{aligned} \mathbb{E}\left[\exp\left(-\int_t^T r(u) du\right) \mid \mathcal{F}_t\right] &= \exp\left(-(T-t)r(t) - \int_t^T \int_t^u \mu(h) dh du + \sigma^2(T-t)^3/6\right) \\ &= \exp\left(A(t, T)r(t) + B(t, T)\right) \end{aligned}$$

Check agrees with previous formula for bonds using relationship between initial forward curve and the drift μ .

Alternative to derive conditional distribution:

$$r(u) = r(t) + \int_t^u \mu(h) dh + \sigma(B_u^{\mathbb{Q}} - B_t^{\mathbb{Q}})$$

$\Rightarrow$ integrate. We use the fact $B_u^{\mathbb{Q}} - B_t^{\mathbb{Q}} \perp \mathcal{F}_t$.

This is actually a proof of the Markov property for r .

But more prosaically we can now integrate to obtain

$$\int_t^T r(u) du = \dots$$

Special case where

$$f(0, t) = -\sigma^2 t/2 + bt + c$$

Then $\mu(t) \equiv b$, so r is a BM with constant drift under EMM — time homogeneous Markov process. We have

$$A(t, T) = A(T - t) = -(T - t)$$

and

$$B(t, T) = B(T - t) = \frac{\sigma^2(T - t)^3}{6} - \frac{b(T - t)^2}{2}$$

A short rate model is a description of the evolution of the short rate under the EMM from which we can deduce the bond prices using the proposition above. There are several nice short rate models which give bond prices of affine form:

Model	Short rate
Ho-Lee	BM with drift
Hull-White	O-U process (time inhomogeneous)
Vasicek	O-U process (time homogeneous)
C.I.R	Squared O-U (time homogeneous)

There are also multi-factor models in which r is a function of a higher dimensional Markov process rather than Markov itself.

16 Lecture 16: Gaussian HJM and Hull-White Model

Assume we are working with an EMM $\mathbb{Q}$. If $\sigma(t, T)$ is deterministic then (since the market price of risk $\Theta \equiv 0$) $\alpha(t, T)$ is deterministic too. Consequently $(f(t, T); t \leq T)$ is a two-parameter Gaussian process, bond prices have log-normal distributions, and the short rate $(r(t); t \geq 0)$ is a Gaussian process too. Note in such a model interest rates can be negative and bond prices greater than 1.

But r is not necessarily a Markov process. In fact, if r is a Markov process, then it has to be an Ornstein–Uhlenbeck process:

$$dr(t) = (a(t) - b(t)r(t)) dt + \sigma(t) dB^{\mathbb{Q}}(t)$$

where a, b, σ are deterministic functions. What can we say about forward rates and bond prices in such a model?

16.1 First Approach

Because r is both Markov and Gaussian, there exist functions M_0, M_1 and V such that the conditional distribution of $\int_t^T r(u) du$ given $\mathcal{F}_t$ is Gaussian with mean $M_1(t, T) r(t) + M_2(t, T)$ and variance $V(t, T)$ (these can be calculated explicitly but are tedious and complicated). Then using the proposition from the previous lecture and the Gaussian MGF:

$$P(t, T) = \exp(A(t, T) r(t) + B(t, T))$$

with

$$A(t, T) = -M_1(t, T), \quad B(t, T) = -M_0(t, T) + V(t, T)/2.$$

Note A, B will satisfy a pair of ODEs. Then from bond prices we could compute forward rates.

16.2 Second Approach

We have

$$f(t, T) = f(0, T) + \int_0^t \alpha(s, T) ds + \int_0^t \sigma(s, T) dB_s^{\mathbb{Q}}$$

So taking $T = t$:

$$r(t) = f(0, t) + \int_0^t \alpha(s, t) ds + \int_0^t \sigma(s, t) dB_s^{\mathbb{Q}}$$

Warning: the integrand depends on t ; this is *not* a local martingale!!

Compare with the solution to the O–U SDE:

$$r(t) = \Phi(t) \left(r(0) + \int_0^t \frac{a(s)}{\Phi(s)} ds + \int_0^t \frac{\sigma(s)}{\Phi(s)} dB_s^{\mathbb{Q}} \right)$$

with $\Phi(t) = \exp\left(-\int_0^t b(s) ds\right)$. (See PS1 Q1.)

The last terms in the two expressions for $r(t)$ match if and only if

$$\sigma(s, t) = \frac{\sigma(s)}{\Phi(s)} \cdot \Phi(t)$$

so Φ factors the forward volatility.

Then $\alpha(t, T)$ is given for $\sigma(t, T)$ by the HJM drift condition; we see:

$$\underbrace{\begin{pmatrix} \sigma(\cdot, \cdot) \\ \alpha(\cdot, \cdot) \end{pmatrix}}_{\text{forward rate parameters}} \longleftrightarrow \underbrace{\begin{pmatrix} \sigma(\cdot) \\ b(\cdot) \end{pmatrix}}_{\text{short rate parameters}}$$

The third parameter in the O–U process is $a(\cdot)$, and this is determined (having fixed the other two) from the initial forward curve, and vice versa, using

$$\Phi(t) \left(r(0) + \int_0^t \frac{a(s)}{\Phi(s)} ds \right) = f(0, t) + \int_0^t \alpha(s, t) ds.$$

Hull–White is used by the banks; but they often constrain $\sigma(\cdot)$ and $b(\cdot)$ to be constants (or piecewise constant) to avoid overfitting. As we have just seen, for given σ , b , we can find $a(\cdot)$ from the initial forward curve, but how does one choose σ , b ? Calibrate to the market price of exchange-traded options such as cap(lets). Then this calibrated model is used to price OTC options.

17 Lecture 17: Forward Measures and Option Prices in Gaussian HJM Models

Let's assume the market filtration generated by the bond prices is the filtration of the BM $B^{\mathbb{Q}}$, then using the PRP for BM it will follow the market is complete and every (integrable) claim can be replicated. This holds in the Gaussian case with $\sigma(t, T)$ deterministic and nonzero. Prices/fair values are then given by the fundamental pricing formula from Lecture 7:

$$\Pi_t(C) = N_t \mathbb{E}_{\mathbb{Q}^N} \left[\frac{C}{N_T} \mid \mathcal{F}_t \right]$$

where N is the numéraire we choose (some tradeable asset), C is the claim to be settled at time T , $\mathbb{Q}^N$ is the EMM* corresponding to N , i.e. under $\mathbb{Q}^N$, $(P(t, T)/N_t)_{t \leq T}$ is a martingale for each T .¹

Up to now we have taken N to be the money account $\exp\left(\int_0^t r(s) ds\right)$ where r is the short rate, and $\mathbb{Q}$ been the EMM corresponding to this so that $\exp\left(-\int_0^t r(s) ds\right) P(t, T)$ is a $\mathbb{Q}$ -martingale. To value C we therefore need to calculate

$$\mathbb{E}_{\mathbb{Q}} \left[\exp\left(-\int_t^T r(s) ds\right) C \mid \mathcal{F}_t \right].$$

When r is random this can be difficult to do directly. Another natural numéraire is the T -bond itself: $N_t = P(t, T)$ for $t < T$, and noting $P(T, T) = 1$ we can write

$$\Pi_t(C) = P(t, T) \cdot \mathbb{E}_{\mathbb{Q}^T}[C \mid \mathcal{F}_t]$$

where $\mathbb{Q}^T$ is the corresponding EMM, usually called the **T -forward measure**. See problem sheet 2.

We need to describe $\mathbb{Q}^T$. We have

$$\mathbb{Q}^T|_{\mathcal{F}_t} = D_t \cdot \mathbb{Q}|_{\mathcal{F}_t} \quad (t \leq T; \text{ note we don't define } \mathbb{Q}^T|_{\mathcal{F}_t} \text{ for } t > T)$$

for some $\mathbb{Q}$ -martingale D . Comparing the expression for $\Pi_0(C)$ to see

$$D_T = \frac{\exp\left(-\int_0^T r(s) ds\right)}{P(0, T)}$$

and hence

$$D_t = \exp\left(-\int_0^t r(s) ds\right) \frac{P(t, T)}{P(0, T)}.$$

Next we want to understand the bond evolution under $\mathbb{Q}^T$. In the HJM framework we have

$$d_t P(t, T) = P(t, T) \left(r(t) dt + \sigma^*(t, T) dB_t^{\mathbb{Q}} \right)$$

with $\sigma^*(t, T) = -\int_t^T \sigma(t, u) du$, with $\sigma(\cdot, \cdot)$ being the forward rate vol (see Lecture 14).

By Girsanov,

$$B_t^T = B_t^{\mathbb{Q}} - \int_0^t \frac{d\langle B^{\mathbb{Q}}, D \rangle(s)}{D(s)} = B_t^{\mathbb{Q}} - \int_0^t \sigma^*(s, T) ds$$

is a $\mathbb{Q}^T$ -BM. Using this, I can re-write the SDE for $P(t, T)$ with B^T .

¹I am just going to assume an EMM (and not just an ELMM) exists.

More usefully: a ratio of bond prices

$$R(t, T, T') = \frac{P(t, T')}{P(t, T)}, \quad t < T \wedge T' \quad \text{is a } \mathbb{Q}^T\text{-martingale by the EMM property of } \mathbb{Q}^T.$$

$$\begin{aligned} \text{And} \quad d_t R &= \frac{d_t P(t, T')}{P(t, T)} - \frac{P(t, T')}{(P(t, T))^2} d_t P(t, T) + \text{further f.v. terms} \\ &= R(\sigma^*(t, T') - \sigma^*(t, T)) dB_t^T \end{aligned}$$

It is also interesting to see what happens to the evolution of the forward rate $f(t, T)$:

$$df(t, T) = \underbrace{-\sigma(t, T) \sigma^*(t, T) dt}_{\text{HJM drift}} + \sigma(t, T) dB_t^{\mathbb{Q}} = \sigma(t, T) dB_t^T,$$

and providing this is a true martingale we deduce

$$\mathbb{E}_{\mathbb{Q}^T}[r(T) | \mathcal{F}_t] = \mathbb{E}_{\mathbb{Q}^T}[f(T, T) | \mathcal{F}_t] = f(t, T),$$

i.e. the forward rate is the expected value of the future short rate under the forward measure.

Lecture 17(b): Forward measures and option prices in the Gaussian HJM model

17.1 Pricing of Claims

We turn to the pricing of claims. We use the following lemma which has an elementary proof; see also PS1 Q2.

Lemma. If $Z \sim N(0, 1)$ then

$$\mathbb{E}\left[(A \exp(\sqrt{v} Z - v/2) - B)^+\right] = A \Phi(d_+) - B \Phi(d_-)$$

$$\text{where } d_{\pm} = \frac{1}{\sqrt{v}} \log(A/B) \pm \frac{1}{2} \sqrt{v}.$$

We use this to price a call option on a bond:

$$C = (P(T, u) - K)^+$$

with $u > T$ and the option settled at time T . Pricing this using the T -forward measure we have:

$$\Pi_0(\text{call}) = P(0, T) \mathbb{E}_{\mathbb{Q}^T}[(P(T, u) - K)^+].$$

Now using our previous result on the evolution of the bond price ratio $R(t, T, u)$ we obtain

$$P(T, u) = \frac{P(T, u)}{P(T, T)} = \frac{P(0, u)}{P(0, T)} \exp\left(\int_0^T (\sigma^*(t, u) - \sigma^*(t, T)) dB_t^T - \frac{1}{2} \int_0^T (\sigma^*(t, u) - \sigma^*(t, T))^2 dt\right).$$

So with deterministic σ^* in the Gaussian HJM model we see $P(T, u)$ has a lognormal distribution under $\mathbb{Q}^T$.

So we may apply the lemma to obtain

$$\Pi_0(\text{call}) = P(0, u) \Phi(d_+) - K P(0, T) \Phi(d_-)$$

where $P(0, u)$ is the initial value of the u -bond and $K P(0, T)$ is the present value of the strike, with

$$d_{\pm} = \frac{1}{\sqrt{v}} \log\left(\frac{P(0, u)}{K P(0, T)}\right) \pm \frac{1}{2} \sqrt{v}, \quad \text{with} \quad v = \int_0^T (\sigma^*(t, u) - \sigma^*(t, T))^2 dt.$$

Essentially the Black-Scholes formula with the u -bond as the underlying.

17.2 Caplets and LIBOR

We turn to another important contract: a one-period cap (or caplet). This pays at time u

$$\kappa (L(T, T, u) - \varkappa)^+$$

where $\kappa = (u - T)$ is the accrual period, T is the set date, u is the payment date, and

$$L(T, T, u) = \frac{1}{u - T} \left(\frac{1}{P(T, u)} - 1 \right)$$

is the (spot) LIBOR.

Three questions about LIBOR:

1. *Why are we studying it if it's been abolished?* We aren't; we are studying simple interest rates.
2. *Does using simple interest rates affect the bonds?* No, the bonds, short rate, forward rate etc. are all as before. We just calculate simple interest rates from those bonds.
3. *Why are we using simple interest rates then?* Because they are used in the market; payments on loans are linear functions of simple rates which makes the maths nice.

$$\Pi_0(\text{caplet}) = P(0, u) \mathbb{E}_{\mathbb{Q}^u} \left[\kappa (L(T, T, u) - \varkappa)^+ \right] = P(0, u) \mathbb{E}_{\mathbb{Q}^u} \left[\left(\frac{1}{P(T, u)} - (1 + \kappa \varkappa) \right)^+ \right].$$

Note we use $\mathbb{Q}^u$ because the cap pays at time u . Now we apply the result on the evolution of the bond ratio $\mathcal{R}(t, u, T) = P(t, T)/P(t, u)$. (Note that T and u are swapped from the call example.) We obtain

$$\frac{1}{P(T, u)} = \frac{P(T, T)}{P(T, u)} = \frac{P(0, T)}{P(0, u)} \exp \left(\int_0^T (\sigma^*(t, T) - \sigma^*(t, u)) dB_t^u - \frac{1}{2} \int_0^T (\sigma^*(t, T) - \sigma^*(t, u))^2 dt \right)$$

and now using the lemma gives

$$\Pi_0(\text{caplet}) = P(0, T) \Phi(d_+) - P(0, u)(1 + \kappa \varkappa) \Phi(d_-)$$

with

$$d_{\pm} = \frac{1}{\sqrt{v}} \log \left(\frac{P(0, T)}{P(0, u)(1 + \kappa \varkappa)} \right) \pm \frac{1}{2} \sqrt{v}, \quad v = \int_0^T (\sigma^*(t, T) - \sigma^*(t, u))^2 dt.$$

18 Lecture 18: The LIBOR Market Model (BGM Model)

Remember we can lock in the cost of borrowing over the period $[T, U]$ at an earlier time $t < T$ by taking a position in bonds with maturities T and U .

Doing this in order to borrow 1 at time T we repay

$$\frac{P(t, T)}{P(t, U)}$$

at time U and so the simple interest rate paid for this loan is the forward LIBOR

$$L(t, T, U) = \frac{1}{(U - T)} \left(\frac{P(t, T)}{P(t, U)} - 1 \right)$$

Note this is a martingale under the U -forward measure $\mathbb{Q}^U$.

Indeed, by considering the SDE for the rates $R(t, U, T)$ from the previous lecture we have

$$dL(t, T, U) = \alpha^{-1} (1 + \alpha L(t, T, U)) (\sigma^*(t, T) - \sigma^*(t, U)) dB_t^U$$

where $\alpha = U - T$ and B^U is a BM under the U -forward measure.

Let us re-write the SDE for L as

$$dL(t, T, U) = L(t, T, U) \lambda(t, T, U) dB_t^U$$

where

$$\lambda(t, T, U) = \frac{1 + \alpha L(t, T, U) (\sigma^*(t, T) - \sigma^*(t, U))}{\alpha L(t, T, U)}$$

is the LIBOR volatility.

In the Gaussian HJM model σ^* is deterministic, λ is not and under $\mathbb{Q}^U$ $L(t, T, U)$ has a displaced log-normal distribution which may take negative values.

Can we instead choose σ^* random and make λ deterministic? This is (was) an important question because the market was typically using Black's caplet formula to infer volatility from observed prices. This says:

If the LIBOR volatility is deterministic then the value at time 0 of a caplet paying $\alpha (L(T, T, U) - K)^+$ at time U is

$$\begin{aligned} \Pi_0(\text{caplet}) &= \alpha P(0, U) (L(0, T, U) \Phi(d_+) - K \Phi(d_-)) \\ d_{\pm} &= \frac{1}{\sqrt{V}} \log \left(\frac{L(0, T, U)}{K} \right) \pm \frac{1}{2} V \quad V = \int_0^T \lambda(t, T, U)^2 dt \end{aligned}$$

Proof. The fundamental pricing formula gives

$$\Pi_0(\text{caplet}) = \alpha P(0, U) \mathbb{E}_{\mathbb{Q}^U} [(L(T, T, U) - K)^+]$$

and

$$L(T, T, U) = L(0, T, U) \exp \left(\int_0^T \lambda(t, T, U) dB_t^U - \frac{1}{2} \int_0^T \lambda(t, T, U)^2 dt \right).$$

Because λ is deterministic the stochastic exponential has a log-normal distribution and we apply the Lemma from the previous lecture to evaluate the expectation.

Back to our question: can λ actually be deterministic?

No, not simultaneously for all T and U (this would entail the instantaneous forward rates $f(t, T)$ being log-normal which can't happen). But the answer is yes if we restrict the set of T and U we consider. Fix $\alpha > 0$ and let

$$T_i = \alpha i, \quad i = 0, 1, \dots, n.$$

Then we consider LIBORs

$$\{L(t, T_i, T_{i+1}) ; t \leq T_i\}$$

for given deterministic volatilities $\lambda(t, T_i, T_{i+1})$ we want

$$dL(t, T_i, T_{i+1}) = L(t, T_i, T_{i+1}) \lambda(t, T_i, T_{i+1}) dB_t^{T_{i+1}}$$

where $B^{T_{i+1}}$ is a BM under the T_{i+1} forward measure $\mathbb{Q}^{T_{i+1}}$.

To solve this system of SDEs we need the relationship between the different B^{T_i} .

At the beginning of Lecture 17 we saw that

$$B^T = B^{\mathbb{Q}} - \int_0^t \sigma^*(s, T) ds$$

where $B^{\mathbb{Q}}$ is a BM w.r.t. $\mathbb{Q}$ the EMM corresponding to the money account being numéraire.

So we suppose that

$$\begin{aligned} dB_t^{T_{i+1}} &= dB_t^{T_i} + (\sigma^*(t, T_i) - \sigma^*(t, T_{i+1})) dt \\ &= dB_t^{T_i} + \frac{\alpha L(t, T_i, T_{i+1}) \lambda(t, T_i, T_{i+1})}{1 + \alpha L(t, T_i, T_{i+1})} dt \end{aligned} \quad (equ : dB_t^{T_{i+1}})$$

using the relation $(\lambda(t, T, U))$ from earlier in this lecture.

Summing over i gives

$$dB_t^{T_i} = dB_t^{T_n} - \sum_{j=i}^{n-1} \frac{\alpha L(t, T_j, T_{j+1}) \lambda(t, T_j, T_{j+1})}{1 + \alpha L(t, T_j, T_{j+1})} dt$$

and we can now re-write the SDEs for the LIBORs so they all driven by the single BM B^{T_n} .

We can solve the SDEs one by one starting with a BM B^{T_n} , and initial values for the LIBORs.

Firstly

$$L(t, T_{n-1}, T_n) = L(0, T_{n-1}, T_n) \mathcal{E} \left(\int_0^t \lambda(s, T_{n-1}, T_n) dB_s^{T_n} \right) (t)$$

solves the SDE for $L(\cdot, T_{n-1}, T_n)$.

Then using this LIBOR and $(equ : dB_t^{T_{i+1}})$ we can define (obtain) $B^{T_{n-1}}$. Then we set

$$L(t, T_{n-2}, T_{n-1}) = L(0, T_{n-2}, T_{n-1}) \mathcal{E} \left(\int_0^t \lambda(s, T_{n-2}, T_{n-1}) dB_s^{T_{n-1}} \right) (t)$$

and continue in the same way until all the LIBORs are constructed.

⇒ Shows the LIBORs form a multidimensional Markov process / diffusion.

⇒ LIBORs give the ratios of bond prices - sufficient for pricing many derivatives. They do not uniquely determine the bond prices, but consistent bond prices can be shown to exist.

19 Lecture 19: Modelling Credit Risk – Default Times

A **default time** is when a creditor fails to repay debt or becomes unable to pay future debt.

Structural Models : Model the internal structure of the creditor; default occurs when assets are insufficient in some way. Assets are modelled by cts processes and default times are predictable.

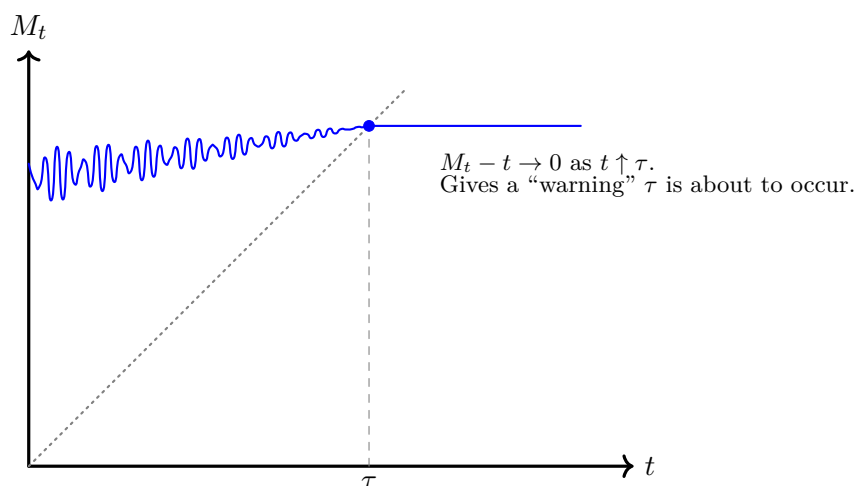
Intensity based Models At a time t , conditional on default not yet having happened, there is a small probability $\lambda(t) dt$ of it occurring in the small time interval $[t, t + dt]$. $\lambda(t)$ is called the hazard rate and may be a function of random factors. Default times are not predictable in such models.

Predictable? Illustration of concept without giving a precise definition.

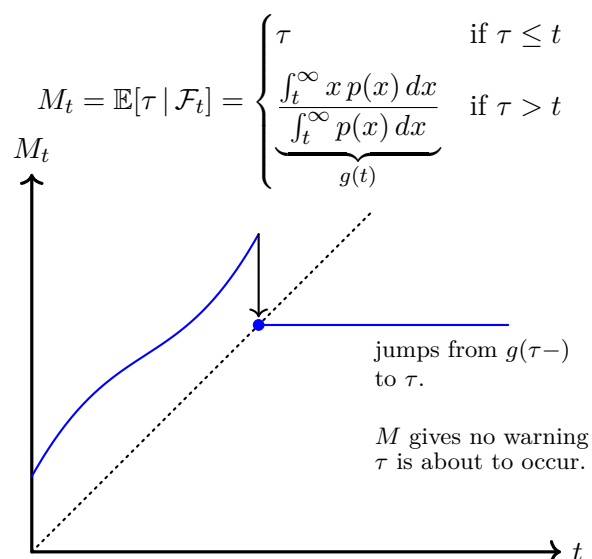
(1) Suppose B is a BM and $(\mathcal{F}_t; t \geq 0)$ the filtration it generates. τ a stopping time with $\mathbb{E} \tau < \infty$. Then

$$M_t = \mathbb{E}[\tau | \mathcal{F}_t]$$

is a continuous martingale by the PRP.



(2) Suppose τ is a +ve random variable whose distribution has a density p . Let $\mathcal{F}_t = \sigma(\{\tau \leq s\}; s \leq t)$, i.e. you know the value of τ once it has occurred but not before. τ is a stopping time w.r.t. this filtration. Assume $\mathbb{E}[\tau] < \infty$.



Important special case: $\lambda > 0$ a constant

$$p(x) = A e^{-\lambda x}$$

so τ has exponential dist with rate A . Then $g(t) = t + 1/\lambda$ because, given τ hasn't yet occurred, the dist of remaining waiting time is $\exp(\lambda)$. This is the **renewal property** of the exp. dist. Observe the crucial difference between (1) and (2) is not the dist of τ ; it is the filtration: "how much you know".

For τ having a general, **strictly +ve density**, we define the **hazard rate**:

$$\lambda(t) = \frac{p(t)}{\int_t^\infty p(x) dx} = \lim_{h \downarrow 0} \frac{1}{h} \mathbb{P}(\tau \in [t, t+h] \mid \tau > t) = -\frac{d}{dt} \log \mathbb{P}(\tau > t) \quad \text{if } p \text{ is cts.}$$

We can recover the distribution of τ from λ :

$$\mathbb{P}(\tau > t) = \exp\left(-\int_0^t \lambda(s) ds\right).$$

And more generally: $\forall t < T$

$$\mathbb{P}(\tau > T \mid \tau > t) = \exp\left(-\int_t^T \lambda(u) du\right)$$

So if $\mathcal{F}_t = \sigma(\{\tau \leq s\}; s \leq t)$ then

$$\mathbb{P}(\tau > T \mid \mathcal{F}_t) = \mathbf{1}_{(\tau > t)} \exp\left(-\int_t^T \lambda(u) du\right).$$

Notice if $\tau \sim \exp(\lambda)$ then the hazard function $\lambda(\cdot) \equiv \lambda$ is constant. In fact we can construct a general τ by time-changing an $\exp(1)$ distributed r.v. e : let $H(t) = \int_0^t \lambda(s) ds$ — cumulative hazard. Then

$$\tau = H^{-1}(e) = \inf\left\{t \geq 0 : \int_0^t \lambda(s) ds \geq e\right\}$$

has hazard rate λ .

Another look at $M_t = \mathbb{E}[\tau \mid \mathcal{F}_t]$. We defined

$$g(t) = \mathbb{E}[\tau \mid \tau > t] = \frac{\int_t^\infty x p(x) dx}{\int_t^\infty p(x) dx}.$$

So

$$\begin{aligned} g'(t) &= -t p(t) / \int_t^\infty p(x) dx + p(t) \int_t^\infty x p(x) dx / \left(\int_t^\infty p(x) dx\right)^2 \\ &= (g(t) - t) \lambda(t) \end{aligned}$$

Now we can see the martingale property of M , informally: Conditional on τ not having occurred by time t , between the times t and $t + dt$

$$M_t \text{ either } \begin{cases} \text{increases by } g'(t) dt & \text{with probability } 1 - \lambda(t) dt \\ \text{decreases by } g(t) - t & \text{with probability } \lambda(t) dt \end{cases}$$

and

$$g'(t) dt \times (1 - \lambda(t) dt) = (g(t) - t) \lambda(t) dt$$

neglecting $(dt)^2$. So two possibilities are *balanced*.

M is a finite variation process but also a (non-constant) martingale. This is only possible because it does not have cts paths.

20 Lecture 20: Doubly Stochastic Times

Idea: Models for τ with the hazard rate λ being random itself.

20.1 Conditional Independence

σ -algebras $\mathcal{A}, \mathcal{B}, \mathcal{C}$.

Defn:

$$\begin{aligned}\mathbb{P}(A \cap C | \mathcal{B}) &= \mathbb{P}(A | \mathcal{B}) \mathbb{P}(C | \mathcal{B}) \quad \forall A \in \mathcal{A}, C \in \mathcal{C} \\ \iff \mathbb{P}(C | \sigma(\mathcal{A}, \mathcal{B})) &= \mathbb{P}(C | \mathcal{B}) \quad \forall C \in \mathcal{C} \\ \iff \mathbb{P}(A | \sigma(\mathcal{B}, \mathcal{C})) &= \mathbb{P}(A | \mathcal{B}) \quad \forall A \in \mathcal{A}.\end{aligned}$$

$\mathcal{A} \perp\!\!\!\perp_{\mathcal{B}} \mathcal{C}$: $\mathcal{A}$ is conditionally independent of $\mathcal{C}$ given $\mathcal{B}$.

e.g. ice cream consumption is found to be correlated with drowning, but when conditioned on weather they are independent.

If the conditional distribution of τ given $\mathcal{G}_{\infty}$ has a strictly +ve density as then λ satisfying (2) exists.

Set up: $\mathcal{G}_t \subseteq \mathcal{F}_t$, $\mathcal{F}_t = \mathcal{G}_t \vee \mathcal{H}_t$ with

$$\mathcal{H}_t = \sigma(\{\tau \leq s\}; s \leq t). \quad (1)$$

Hazard rate λ assumed to be adapted to $\mathcal{G}$ and

$$\mathbb{P}(\tau > t | \mathcal{G}_{\infty}) = \exp\left(-\int_0^t \lambda(s) ds\right). \quad (2)$$

Construction: take $e \perp\!\!\!\perp \mathcal{G}_{\infty}$ with $\exp(1)$ distribution and let $\tau = \inf\{t \geq 0 : \int_0^t \lambda(s) ds \geq e\}$.

Proposition

For $t < T$

$$\mathbb{P}(\tau > T | \mathcal{F}_t) = \mathbf{1}_{(\tau > t)} \mathbb{E}\left[\exp\left(-\int_t^T \lambda(u) du\right) | \mathcal{G}_t\right]$$

and more generally, for Y $\mathcal{G}_{\infty}$ -measurable (and integrable)

$$\mathbb{E}[Y \mathbf{1}_{(\tau > T)} | \mathcal{F}_t] = \mathbf{1}_{(\tau > t)} \mathbb{E}\left[Y \exp\left(-\int_t^T \lambda(u) du\right) | \mathcal{G}_t\right].$$

Note: this generalises the formula for $\mathbb{P}(\tau > T | \mathcal{F}_t)$ given in the last lecture for the case where λ was a deterministic function.

20.2 Example:

$(\mathcal{G}_t)_{t \geq 0}$ generated by a BM B and λ satisfies the SDE

$$d\lambda(t) = (a - b\lambda(t)) dt + \sigma\sqrt{\lambda(t)} dB(t)$$

$a > 0 \Rightarrow \lambda(t) > 0 \forall t$ (same diffusion as in Heston's model + CIR).

Then we have

$$\begin{aligned}\mathbb{P}(\tau > T | \mathcal{F}_t) &= \mathbf{1}_{(\tau > t)} \mathbb{E}\left[\exp\left(-\int_t^T \lambda(u) du\right) | \mathcal{G}_t\right] \\ &= \mathbf{1}_{(\tau > t)} \exp(A(T-t)\lambda(t) + B(T-t))\end{aligned}$$

for deterministic functions A, B - exactly the same as in the interest rate model PS5 Q2.

Proof of the Proposition

In the set-up described above $\mathcal{F}_t \perp\!\!\!\perp_{\mathcal{G}_t} \mathcal{G}_\infty$.

To see this, first observe that $\mathcal{F}_t$ is, by defn, generated by events $\{\tau \leq s\} \cap A$, $A \in \mathcal{G}_t$, $s \leq t$. Then,

$$\mathbb{P}(\{\tau \leq s\} \cap A \mid \mathcal{G}_\infty) = \mathbf{1}_A \mathbb{P}(\tau \leq s \mid \mathcal{G}_\infty) = \mathbf{1}_A \left(1 - \exp\left(-\int_0^s \lambda(h) dh\right)\right).$$

But the R.H.S. is $\mathcal{G}_t$ -measurable and so

$$\mathbb{P}(\{\tau \leq s\} \cap A \mid \mathcal{G}_t) = \mathbb{P}(\{\tau \leq s\} \cap A \mid \mathcal{G}_\infty),$$

which implies the conditional independence.

Note: quite natural property. The future evolution of underlying factors is not affected by whether τ has occurred yet.

We now proceed to the proof of the identity stated in the proposition. Because $\mathcal{H}_t = \sigma(\{\tau \leq s\}; s \leq t)$ we have

$$\mathbb{P}(\tau > T \mid \mathcal{G}_\infty \vee \mathcal{H}_t) = \begin{cases} 0 & \text{if } \tau \leq t, \\ \frac{\mathbb{P}(\tau > T \mid \mathcal{G}_\infty)}{\mathbb{P}(\tau > t \mid \mathcal{G}_\infty)} & \text{if } \tau > t, \end{cases}$$

and so

$$\mathbb{P}(\tau > T \mid \mathcal{G}_\infty \vee \mathcal{H}_t) = \mathbf{1}_{(\tau > t)} \exp\left(-\int_t^T \lambda(u) du\right).$$

Next, because $\mathcal{F}_t \subset \mathcal{G}_\infty \vee \mathcal{H}_t$, and Y is $\mathcal{G}_\infty$ -measurable

$$\begin{aligned} \mathbb{E}[Y \mathbf{1}_{(\tau > T)} \mid \mathcal{F}_t] &= \mathbb{E}[\mathbb{E}[Y \mathbf{1}_{(\tau > T)} \mid \mathcal{G}_\infty \vee \mathcal{H}_t] \mid \mathcal{F}_t] \\ &= \mathbb{E}[Y \mathbb{P}(\tau > T \mid \mathcal{G}_\infty \vee \mathcal{H}_t) \mid \mathcal{F}_t] \\ &= \mathbb{E}\left[Y \mathbf{1}_{(\tau > t)} \exp\left(-\int_t^T \lambda(u) du\right) \mid \mathcal{F}_t\right] \\ &= \mathbf{1}_{(\tau > t)} \mathbb{E}\left[Y \exp\left(-\int_t^T \lambda(u) du\right) \mid \mathcal{F}_t\right] \\ &= \mathbf{1}_{(\tau > t)} \mathbb{E}\left[Y \exp\left(-\int_t^T \lambda(u) du\right) \mid \mathcal{G}_t\right] \end{aligned}$$

where the last equality follows by the conditional independence.

21 Lecture 21: Bonds with Default

We consider a financial market consisting of bonds with price $P(t, T)$ which are guaranteed to pay unity at maturity, and also bonds subject to default with prices $\hat{P}(t, T)$ paying unity at maturity T if $\tau > T$ and otherwise 0. Here τ is a r.v. representing the default time, the same r.v. for all T . We have $\hat{P}(T, T) = \mathbf{1}_{(\tau > T)}$.

There is also a money account with short rate $r(t)$. We let $(\mathcal{F}_t)_{t \geq 0}$ be the filtration representing the information available in the market and suppose $\mathbb{Q}$ is an EMM with the numéraire being the money account: $\exp\left(\int_0^t r(s) ds\right)$.

We assume the set-up of the last lecture so

$$\mathcal{F}_t = \mathcal{G}_t \vee \mathcal{H}_t$$

where $\mathcal{G}_t \subseteq \mathcal{F}_t$, $\mathcal{H}_t = \sigma(\{\tau \leq s\}; s \leq t)$, with

$$\mathbb{Q}(\tau > t | \mathcal{G}_\infty) = \exp\left(-\int_0^t \lambda(s) ds\right)$$

and the hazard rate λ and short rate r both being $\mathcal{G}_t$ -adapted. What can we say about the bond prices P and $\hat{P}$?

By defn of $\mathbb{Q}$ as an EMM, the discounted bonds are martingales w.r.t. $(\mathcal{F}_t)_{t \geq 0}$, i.e.

$$\exp\left(-\int_0^t r(s) ds\right) P(t, T) \quad \text{are martingales.}$$

$$\exp\left(-\int_0^t r(s) ds\right) \hat{P}(t, T)$$

Since $P(T, T) = 1$ we can deduce from the martingale property that

$$\begin{aligned} P(t, T) &= \mathbb{E}_{\mathbb{Q}} \left[\exp\left(-\int_t^T r(u) du\right) \mid \mathcal{F}_t \right] \\ &= \mathbb{E}_{\mathbb{Q}} \left[\exp\left(-\int_t^T r(u) du\right) \mid \mathcal{G}_t \right] \end{aligned}$$

because $\mathcal{F}_t \perp\!\!\!\perp_{\mathcal{G}_t} \mathcal{G}_\infty$. Similarly as $\hat{P}(T, T) = \mathbf{1}_{(\tau > T)}$ the martingale property gives

$$\begin{aligned} \hat{P}(t, T) &= \mathbb{E}_{\mathbb{Q}} \left[\mathbf{1}_{(\tau > T)} \exp\left(-\int_t^T r(u) du\right) \mid \mathcal{F}_t \right] \\ &= \mathbf{1}_{(\tau > t)} \mathbb{E}_{\mathbb{Q}} \left[\exp\left(-\int_t^T (r(u) + \lambda(u)) du\right) \mid \mathcal{G}_t \right] \end{aligned}$$

by Proposition of last lecture. So the effect of the possibility of default is that the bond is priced as if the interest rate was increased by the hazard rate. But note this is the hazard rate under $\mathbb{Q}$, not the real world hazard rate.

21.1 Examples

(1) λ deterministic then

$$\begin{aligned} \hat{P}(t, T) &= \mathbf{1}_{(\tau > t)} \exp\left(-\int_t^T \lambda(u) du\right) \times \mathbb{E} \left[\exp\left(-\int_t^T r(u) du\right) \mid \mathcal{G}_t \right] \\ &= \mathbf{1}_{(\tau > t)} \mathbb{Q}(\tau > T | \tau > t) P(t, T). \end{aligned}$$

(2) C.I.R. model for $r(\cdot)$

$$dr(t) = (a - b r(t)) dt + \sigma \sqrt{r(t)} dB(t)$$

and $\lambda(t) = c_1 r(t) + c_0$, c_0, c_1 constants.

$\mathcal{G}$ generated by the BM B .

Then

$$P(t, T) = \exp(-A(T-t)r(t) - B(T-t))$$

where A, B solve the ODEs

$$\begin{aligned} A'(t) &= 1 - \frac{\sigma^2}{2} (A(t))^2 - b A(t) \\ B'(t) &= a A(t). \end{aligned}$$

And

$$\begin{aligned} \hat{P}(t, T) &= \mathbf{1}_{(\tau > t)} \mathbb{E}_{\mathbb{Q}} \left[\exp \left(- \int_t^T (r(u) + \lambda(u)) du \right) \mid \mathcal{G}_t \right] \\ &= \mathbf{1}_{(\tau > t)} \mathbb{E}_{\mathbb{Q}} \left[\exp \left(-(c_1 + 1) \int_t^T r(u) du - c_0(T-t) \right) \mid \mathcal{G}_t \right] \\ &= \mathbf{1}_{(\tau > t)} \exp(-\tilde{A}(T-t)r(t) - \tilde{B}(T-t)) \end{aligned}$$

where $\tilde{A}, \tilde{B}$ solve the ODEs:

$$\begin{aligned} \tilde{A}'(t) &= (c_1 + 1) - \frac{\sigma^2}{2} (\tilde{A}(t))^2 - b \tilde{A}(t) \\ \tilde{B}'(t) &= c_0 + a \tilde{A}(t), \end{aligned}$$

by generalising PS5 Q2 to compute the conditional expectation. (**Note:** could use stochastic calculus for general processes to check $\hat{P}$ is a martingale). How is a model like (2) used? Parameters are chosen by fitting to current prices of bonds / other instruments (written on bonds). Then the model can be used to price more complicated contracts.

Important example of another instrument used: **credit default swaps**. These work as insurance policies:

- premium leg: cts payment at rate K until default τ or expiry time T .
- protection leg: payment 1 made at default time if it occurs before expiry time T .

Value at time 0:

$$\mathbb{E}_{\mathbb{Q}} \left[\exp \left(- \int_0^T r(s) ds \right) \mathbf{1}_{(\tau \leq T)} - K \int_0^{T \wedge \tau} \exp \left(- \int_0^t r(s) ds \right) dt \right]$$

$\mathbb{Q}$ being the EMM as before. Why? Need to replicate the cashflow using bonds.

For the model used in (2) above based on CIR we can evaluate this expression using ODEs for $\tilde{A}, \tilde{B}$.

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